Aladdin Model Documentation

BlackRock Fundamental Risk for Equities (BFRE)

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This paper describes a suite of fundamental equity risk models estimated and maintained by BlackRock. These are collectively known as BlackRock Fundamental Risk for Equities, or BFRE (pronounced B-Free). The purpose of these models is to describe the structure and sources of equity portfolio risk globally. This paper covers all versions of the BFRE models namely BFRE 1.0 and 2.0.

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1 Introduction

This paper describes a suite of fundamental equity risk models estimated and maintained by BlackRock. These are collectively known as BlackRock Fundamental Risk for Equities, or BFRE (pronounced B-Free). The purpose of these models is to describe the structure and sources of equity portfolio risk globally.

The distinguishing feature of these models is the use of company fundamentals, in addition to historical market data, to describe the risk profile of equity assets and portfolios. This is in contrast to other types of equity risk models that rely solely on historical stock returns to generate ex-ante forecasts of portfolio risk. Forecasts based on stock returns alone can quickly become misleading if a company undergoes changes in its operating activities, or is subject to corporate actions, or experiences changes to its capital structure. These changes would be immediately reflected in the risk forecasts provided by a fundamental equity risk model, whereas they would only gradually be incorporated into risk forecasts based solely on stock returns. There is a long history of using fundamental information to forecast portfolio risk in this manner. The approach here draws on the pioneering work of Rosenberg [1] and the related literature, and combines lessons learned over several decades of academic research, with new insights from investment and risk professionals at BlackRock and Aladdin Financial Engineering (AFE).

The BFRE models are factor-based models that can be used to decompose portfolio risk into a series of factors based on technical and fundamental characteristics. Many of these characteristics are similar to those used by portfolio managers and financial analysts to form investment views on the future performance of companies. These characteristics are based on balance sheet variables, income statements, analyst estimates, country and industry of business, and historical market data. One of the primary advantages of using a fundamental equity risk model is the decomposition of risk using an intuitive set of fundamental equity risk factors that align with how portfolio managers structure their investments. This paper covers BFRE models including both BFRE 1.0 models and BFRE 2.0 enhancements.

The BFRE 2.0 models are enhanced to produce risk forecasts that better respond to latest market conditions. The Short Horizon 2.0 models address the challenge of rapid shifts in recent volatility regime and are tailored to a 1-month forecast horizon. The Long Horizon 2.0 models adapt to shifts in long term volatility and are tailored to a 6-month horizon.

The BFRE models extend Aladdin's equity risk modelling capabilities by providing an alternative and entirely complementary methodology to the one used in Aladdin's Security Total Return Methodology, also known as STORM. In STORM each asset is effectively treated as its own factor, and portfolio risk is calculated using an asset-by-asset covariance matrix estimated from asset returns alone. The alternative approach taken in BFRE imposes far more structure on the asset covariance matrix, reducing the modelling problem to a smaller set of factors, which capture the most important sources of asset return commonality. In doing so, this reframes the risk forecasting problem into one of deciding what factors influence cross-sectional equity returns, and then deciding how best to forecast the risk of these common structural sources.

The current suite of BFRE 1.0 models is made up of five regional models covering North America; Europe, Middle East & Africa; Asia Pacific excluding Japan; Latin America & Caribbean; and Emerging Markets, nine country models for Japan; China; South Africa; Brazil; United Kingdom; Australia; Canada; United States; and India, a sector-specific model covering North American Health Care, and a World model covering the superset of all assets in the regional and country models. The current set of BFRE 2.0 models released include BFRE 2.0 Short Horizon North America, Long Horizon North America, Short Horizon World and Long Horizon World. The suite of BFRE 2.0 models will be extended to other regions and countries in the future. These can be accessed through various Aladdin applications including Portfolio Risk Tools (PRT), Explore, Portfolio Construction (PfC), and the Green Package.

2 Executive Summary

- BFRE is a suite of fundamental equity risk models. BFRE 1.0 is currently made up of five regional models covering North America; Europe, Middle East & Africa; Asia Pacific excluding Japan; Latin America & Caribbean; and Emerging Markets, nine country models for Japan; China; South Africa; Brazil; United Kingdom; Australia; Canada; United States; and India, a sector-specific model covering North American Health Care, and a World model covering the superset of all assets in the regional and country models. The current set of BFRE 2.0 models released include BFRE 2.0 Short Horizon North America, Long Horizon North America, Short Horizon World and Long Horizon World. The suite of BFRE 2.0 models will be extended to other regions and countries in the future
- BFRE provides asset coverage for 87[1] countries spanning all major exchanges and equity asset types: common shares, ordinary shares, preference shares, ADRs, GDRs, certificates, investment trusts and closed-end funds
- BFRE model frequency for factor covariance and specific risk forecast is daily for all the models except WRLD and EMKT
- The model structure includes a market factor, style factors, industry factors, country factors and currency factors; style and industry factors vary somewhat by region reflecting local variation in the importance of different factors
- Industry factors are derived from third-party industry classification schemas[2], and reflect statistical significance
- Style factors are based on well-known and long established investment trends, e.g. value, growth, large-cap vs. small-cap, and pervasive macro-economic themes, e.g. oil price dynamics, emerging markets vs. developed markets
- Model estimation is performed in cross-section in two-passes with precedence on more representative assets in countries and industries with higher development status and better data coverage and quality
- Factor covariance matrices are constructed using a model frequency time-series of Factor Returns correcting for serial correlations
- Specific risk forecasts are based on a time-series of asset-level specific returns whose frequency aligns with that of the model
- The factor covariance and specific risk forecasts are enhanced in BFRE 2.0 versions of the risk model so that they dynamically adapt to current market conditions
- Specific return correlations capture linkages between specific risks of related assets in the same company, e.g. between ordinary shares and preferences shares
- Monthly updates to model estimation universes; weekly updates to factor exposures[3] and specific risk forecasts, and daily updates to model coverage universes (to reflect corporate actions) and factor covariance matrices; daily Factor Returns

[1] BFRE 2.0 covers 86 countries excluding Venezuela from coverage
[2] The Refinitiv Business Classification for all models
[3] more recent models including BFRE 1.0 India, China, North American Healthcare, South Africa, Brazil, and BFRE 2.0 models have daily updates, research history for older models has monthly updates

3 Methodology

3.1 Model Structure

The identification of a relevant set of factors is a critical part of the model construction process of any factor-based approach. The BFRE models use common factors based on highly intuitive, technical and fundamental security characteristics. All common factors are selected based on the following criteria:

- Interpretability: factors should be easily interpretable, and have a strong economic rationale;
- Explanatory Power: factors should explain the covariance structure of stock returns;
- Consistency: factors should be significant through time across different market regimes; and
- Efficacy: factors should improve forecasts of portfolio beta and portfolio risk, and explain portfolio performance in factor-based attribution

Each equity, and by extension, all portfolios that contain equities have exposures to these common factors. The common factors used in BFRE can be categorised into five blocks: a market factor, country factors, currency factors, industry factors and style factors. These are described in more detail below.

Market factor

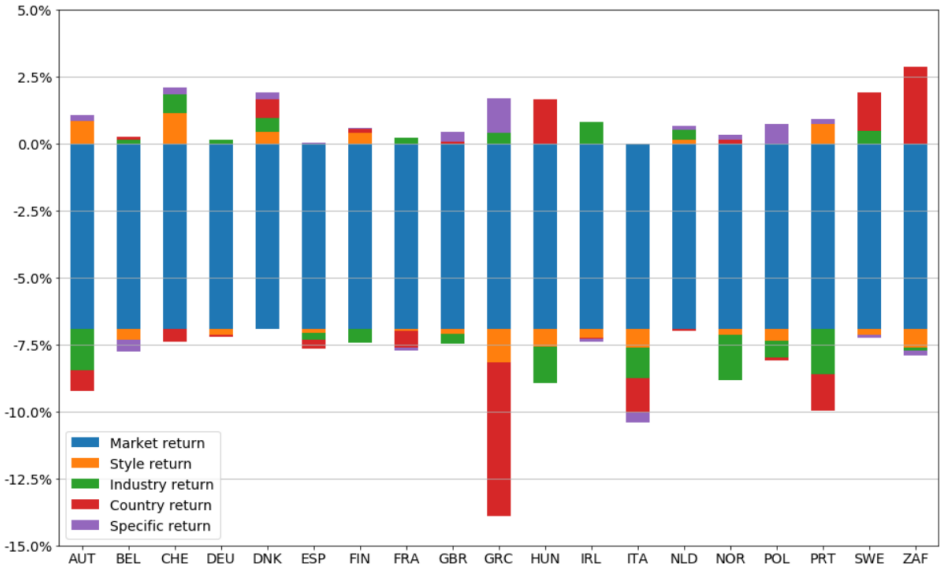
All BFRE models are specified with a market factor and all equity assets have a unit exposure to this factor. The Market factor return represents the cross-sectional average[4] return across all assets in the model estimation, and captures the risk associated with general market movements. The return to the Market factor is highly correlated but not identical to the return to a comparable market index covering the same region. For example, a regression of daily S&P 500 excess returns on daily NAMR Market factor returns yields a beta of 0.99 with an R2 of 91%. This is not expected to be a perfect fit as the NAMR market covers a broader universe of assets, including Canadian stocks, and uses square-root of market capitalisation weights, which assigns greater weight to the mid-cap and small-cap segments relative to large-cap stocks. Nevertheless, both are expected to be highly correlated through time and exhibit similar levels of volatility.

The Market factor exposure of a portfolio should not be confused with its market beta. The Market factor exposure represents the fraction of portfolio %NAV invested in equities[5]. The BFRE model does not include a single, explicit portfolio market beta exposure. Portfolio beta to a market index can instead be computed via the risk factor exposures of the portfolio and market index, together with the factor covariance matrix.

The addition of a market factor in the model specification leads to a cleaner separation and interpretation of market, industry and country effects, and is a standard approach adopted in the academic literature[6]. This is especially useful in risk decomposition and factor attribution analysis and provides a better understanding of the contribution of different effects on portfolio risk and return. As an example, consider the performance of some European country indices[7] on 9 March 2020. Markets across countries in Europe faced large drawdowns, as they dealt with the the start the of COVID-19 pandemic. Figure 1 shows a factor attribution for each index giving the contribution to return from each factor block in the model[8]. The largest contribution to return is from the Market factor, which captures the average (negative) return across stocks on this day. The magnitude is uniform across each index as all are fully invested and so have a Market factor exposure of one. Additionally, there are smaller, varying contributions from style, industry, country and specific returns; however, the dominant source of return is clearly due to a broad weakness across markets, which is captured

[4]Average return based on regression weights, i.e. square-root of market capitalisation
[5]and delta-adjusted market exposure for derivatives
[6]Heston and Rouwenhorst [2]
[7]Indices are market capitalisation weighted, country carve-outs of the EMEA estimation universe
[8]No currency exposure is assumed

Figure 1: factor attribution of country indices in the EMEA model, 9 March 2020



by the Market factor. Contrast this with Figure 2, which shows a similar factor attribution on 30 September 2020, which represents a more benign market environment where sector and geographical factors drive the dispersion in returns and the Market factor contribution is no longer the largest source of return in the attribution. In this instance, any large country index returns are largely attributed to the country factor. This can be seen clearly for Greece (GRC) and Italy (ITA). Given these country indices generally represent well diversified portfolios, specific returns are for the most part negligible.

Country factors

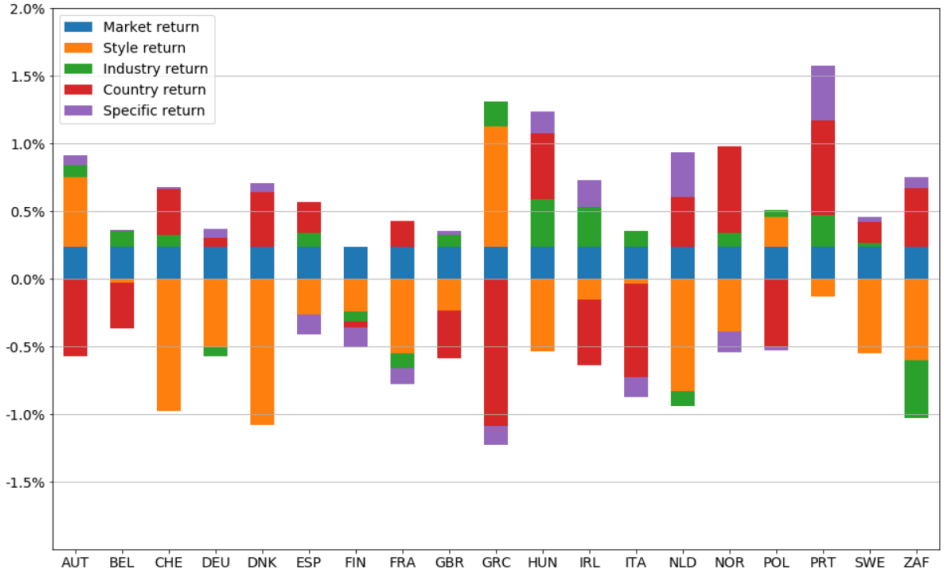
Country factors are only included in the specification of all regional models (i.e. models spanning multiple countries). Models covering a single country, such as Japan, do not require a separate country factor as this role is assumed by the Market factor. Country Factor Returns are estimated in a multivariate regression together with other common factors, and are therefore adjusted to be neutral with respect to market, style and industry effects. This separation allows us to disentangle related (and often correlated) effects, e.g. a country with a large concentration in a single industry, so that model users can understand the risks associated with bearing country exposure and industry exposure in isolation. Every asset is assigned a unit exposure to a single country factor at any point in time.

Country factor exposures are assigned based on a company's country of business[9]. In some special cases, the country of business is a minor region or tax jurisdiction, which is not significant enough to warrant its own factor in the model. In these instances, the country of business will be mapped to an appropriate proxy country, e.g. assets in Panama are reclassified to have a country exposure of USA.

The model coverage universe is defined to include all listings trading within a given region, including listings of companies which are foreign to a particular region. In these cases, all listings from a foreign company would be assigned a country factor exposure based on the country of listing of its most liquid listing, e.g. the listings of AT&T in the EMEA model will all be assigned to Germany, which is the country of the most liquid listing of AT&T within the EMEA model (as there is no USA country factor).

[9]Country Exposures are primarily determined using a hierarchy of Worldscope country of business, and Datastream country code

Figure 2: factor attribution of country indices in the EMEA model, 30 September 2020



Currency factors

The BFRE models are designed to be interpretable from any currency perspective. Each model includes a series of currency factors to handle currency risk. The Factor Returns of non-currency factors (market, styles, industries and countries) are estimated from local asset returns in excess of the local risk free rate, which ensures that they are fully hedged from a currency perspective. The currency Factor Returns are calculated[10] from daily exchange rates and risk-free rates. These returns are expressed consistently with respect to a common numeraire, chosen to be US dollars. This choice can be changed by performing a simple rotation, for example, to the base currency of a non-US portfolio.

Currency factor exposures are assigned in accordance with the country factor exposure using a one-to-one mapping between countries and their principal traded currency. Every asset is assigned a unit exposure to single currency factor at any point in time.

Industry factors

All BFRE industry factors are derived from third party industry schemas (The Refinitiv Business Classification for all models), which provide multiple levels of classification within a nested structure. The lowest (the least granular) level maps the whole market into a small number of sectors while the highest (the most granular) level maps to a large number of sub-industries. All of the industries in Level $n+1$ are sub-industries of some industry in Level n. The third party industry schemas BFRE employs are widely accepted by market participants as an appropriate industry analysis framework for investment research and portfolio management, and as such, serve as a good starting point for the industry factor selection. Note that not all industries at the same level in an industry schema have equal explanatory power. In some cases individual sub-industries may be economically important and statistically significant, justifying their own category. In other cases, a sector view may be sufficient. Therefore further research is conducted to determine the appropriate sector cohort whilst ensuring that each grouping contains a reasonable number of assets and represents a non-negligible proportion of market capitalisation.

Industry factors are defined and estimated globally across all countries in the estimation universe. The only exception is the Emerging Market model which uses local industry definitions for certain sectors. Local indus-

[10]Note the use of the term 'calculated' here, as opposed to 'estimated' – no uncertainty is involved in their construction

try factors can be estimated at a country- or regional-level, and are designed to capture differences in asset return dynamics within a given industry. To maintain parsimony in the factor structure, local industry factors are only used in instances where there is a strong economic rationale for local effects, and this is supported by significant statistical evidence. For example, this is an important consideration in the banking industry where locality specific regulations, fiscal and monetary policies give rise to greater heterogeneity in this group of assets. This is in contrast to developed markets, where there is more cohesion between regulatory bodies, and markets are more homogeneous.

Industry factor selection is determined primarily using statistical criteria. BFRE adopts a frequentist stepwise regression approach. We refer the reader to Miller [3], Hocking [4] and Thompson [5] for detailed discussions on frequentist methods. These approaches suggest different heuristics in selecting factors with the most explanatory power from a large pool of candidate factors.

The BFRE industry factor selection proceeds as follows. We first run a cross sectional regression of asset returns on the sectors in the least granular level and then the residuals of that model are regressed against the next, more granular level of the schema to compute the marginal benefit of increasing the granularity. More specifically, for $n = 1, 2, \ldots$, we run the following two-step regression:

$$r = X_{\text{Mkt}} f_{\text{Mkt}} + \sum_{k \in \text{CCty}} X_{\text{CCty},k} f_{\text{CCty},k} + \sum_{j \in \text{n-Ind}} X_{\text{n-Ind},j} f_{\text{n-Ind},j} + u \tag{1}$$

$$u = \sum_{j \in \text{(n+1)-Ind}} X_{\text{(n+1)-Ind},j} f_{\text{(n+1)-Ind},j} + \tilde{u} \tag{2}$$

where

r	: monthly local asset returns in excess of the risk-free rate
$X_{\text{Mkt}} f_{\text{Mkt}}$	: Market exposures and Factor Returns
$X_{\text{CCty},k} f_{\text{CCty},k}$	: core country exposures and factor returns
$X_{\text{n-Ind},j} f_{\text{n-Ind},j}$	: third party industry schema level n industry exposures and factor returns
u	: residual returns from the first step regression
$\tilde{u}$	: residual returns from the second step regression

These multivariate regressions are performed monthly over the 15-year research history (1996-2010) for the first set of BFRE models released in 2013. The analysis is summarised over the full sample period as well as on five-year sub-samples. Individual factor efficacy is assessed using a history of t-statistics. We consider an absolute t-statistics in excess of 2 as statistically significant. The proportion of significant t-statistics for each industry factor serves as a good proxy for the persistence of individual factor effects. Also, to distinguish between factors with t-statistics close to +/- 2 and those that are significantly higher, we compute the average squared t-statistic. Note that a large proportion of significant t-statistics in the second step regression is viewed as evidence that the increased granularity improves the explanatory power of the model. In addition, to ensure that there are sufficient number of firms to calibrate a separate industry factor in our schema, we compute the market capitalization weight and the effective number of assets for each industry at each level.

An industry is eligible to enter our industry schema as a candidate factor if it has a large proportion of significant t-statistics, a high number of average squared t-statistics, a large market capitalization weight and a large effective number of assets. We identify eligible factors by examining industries in the most granular level and work up the industry hierarchy to less granular levels. In instances where not all of the industries at a given level are considered candidate factors, a re-grouping decision is required based upon cluster analysis. Cluster analysis uses return data as an input and suggests a grouping strategy such that industries in the same group are more closely related to each other than to those in other groups.

Style factors

Style factors capture sources of asset return commonality not captured by market, country and industry factors. Many of these styles are well-known investment themes in the asset management industry and measure

exposure to firm attributes such as value, growth and size. These are useful as a company's exposure to such attributes will capture the degree to which its return behaviour is influenced by economic and business cycle risks. In contrast, other style factors in the model take a more direct approach and measure exposure to economic risks using empirical analysis based on historical market data. In both cases there is substantial empirical evidence supporting the use of these style factors in multi-factor risk models of this form. The majority of BFRE style factors feature in the seminal research of Rosenberg [1] and the related literature, and their inclusion in the models signify their continued relevance in explaining global and regional sources of equity risk through different market environments.

Table 1 provides a short description of each style factor, Table 2 shows some summary statistics for the market and style factors in the NAMR model, and Figure 3 shows the cross-sectional style factor exposure correlations for the NAMR model.

Table 1: style factor descriptions

Style	Description
Size	Company size based on market capitalisation & fundamental data
Volatility	Various measures of historical volatility & historical beta
Small-Cap	Small-cap adjustment to the (linear) size factor exposure
Mid-Cap	Mid-cap adjustment to the (linear) size factor exposure
Reversal	One month return capturing short-term overreaction
Momentum	Longer-term trend in stock prices over the last year
Liquidity	Various measures of trading activity & price impact
Value	Identifies cheap vs. expensive stocks relative to fundamentals
Earnings Yield	Earnings-to-price & related measures
Dividend Yield	Dividend-to-price
Profitability	Return on equity (ROE) & related measures
Growth	Historical growth in assets & sales
Leverage	Various measures of indebtedness
Quality	A measure of shareholder dilution
Foreign Sensitivity	Proportion of foreign assets and revenues
Emerging Market	Identifies developed market stocks with emerging market exposure
Oil	Sensitivity of stock return with changes to the oil price
Sentiment	Sensitivity of stock return with changes in the VIX index

Style factors are constructed from company fundamentals, analyst estimate data and historical market data. These are typically defined as an equal weighted combination of substyles, where each substyle reflects a different formulation of the style. As an example, a definition of size may use three different substyles: market capitalisation, total sales and the number of analyst estimates for FY1 EPS[11]. Combining multiple substyles leads a more robust definition of style Factor Returns and typically increases the explanatory power of the factor.

Substyles are standardised to a common scale to facilitate comparison of securities across different regions, and to allow substyles, measured in different units, to be combined to form style factor exposures. Conceptually, the standardisation process is similar to forming a z-score. The mean is defined as the square-root of

[11] Earnings-per-share (EPS) forecasts for the current fiscal year (FY1)

Table 2: market and style factor summary statistics for the NAMR model, Feb 2000 – Dec 2024

	Annualised Return	Annualised Volatility	Sharpe Ratio	Correlation with Market Factor
Market	6.4%	19.6%	0.32	1.00
Size	-1.1%	3.0%	-0.37	0.22
Volatility	-1.5%	7.4%	-0.20	0.79
Mid-cap	0.5%	1.7%	0.30	0.13
Reversal	-4.7%	3.8%	-1.21	-0.41
Momentum	2.0%	4.6%	0.43	-0.10
Liquidity	5.1%	4.8%	1.05	0.65
Value	2.5%	2.3%	1.08	0.09
Earnings Yield	0.2%	2.1%	0.09	0.06
Dividend Yield	-0.8%	1.9%	-0.39	-0.02
Profitability	3.1%	2.2%	1.43	-0.05
Growth	-1.3%	2.1%	-0.64	0.17
Sentiment	-0.2%	1.7%	-0.12	0.03

market capitalisation weighted average value so that the transformed substyles (and styles) have the property that their weighted average is zero. Additionally, these values are divided by their equally-weighted standard deviation so that a value of +1 for a substyle or style can be interpreted as a security having an exposure of one standard deviation above the market average. An exposure of zero indicates that a security has the market average value for a given substyle, or style.

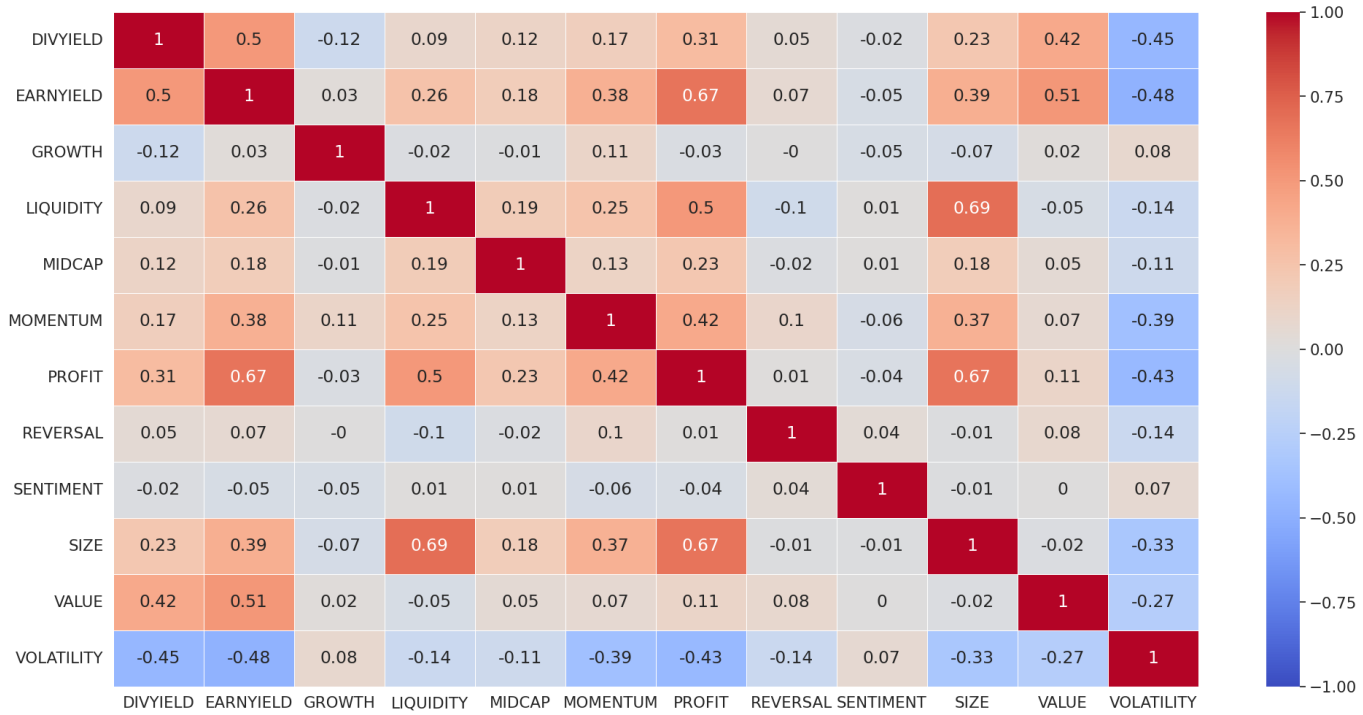
Style factors are added recursively in the model construction process based on their incremental significance. The entire process uses an exhaustive set of over 200 substyles taken from the academic literature and current investment risk practice. The full list of candidate substyles is provided in Table 6. We begin the model construction process with estimating a model specified with market, country and industry factors, and then regress the residuals of this model on each candidate substyle. In this stage, for $i = 1, 2, \cdots, N$ (where $N \geq 200$ is the number of candidate substyles) we run the following two-step regression:

$$r = X_{\text{Mkt}} f_{\text{Mkt}} + \sum_{k \in \text{CCty}} X_{\text{CCty},k} f_{\text{CCty},k} + \sum_{j \in \text{CInd}} X_{\text{CInd},j} f_{\text{CInd},j} + \epsilon \tag{3}$$

$$\epsilon = X_{\text{subSty},i} f_{\text{subSty},i} + \tilde{\epsilon}_i \tag{4}$$

where

Figure 3: style exposure correlations in the NAMR model, Dec 2024



r : monthly local asset returns in excess of the risk-free rate

$X_{\text{Mkt}} f_{\text{Mkt}}$: market exposures and Factor Returns

$X_{\text{CCty},k} f_{\text{CCty},k}$: core country exposures and factor returns

$X_{\text{CInd},j} f_{\text{CInd},j}$: core industry exposures and factor returns

$X_{\text{subSty},i} f_{\text{subSty},i}$: substyle i exposures and factor returns

ϵ : residual returns from the regression on market, core countries and core industries

$\tilde{\epsilon}_i$: residual returns from the regression on factors including substyle i

Candidate substyles are then ranked according to statistical significance metrics over the full sample (and sub-samples). Special emphasis is placed on statistical significance in the last five years of the research history so that emerging stylistic themes are appropriately incorporated into the models. Typically the top-ranking substyles all revolve around a common theme. As an example, Figure 4 shows the top four candidate substyles with the largest proportion of significant t-statistics in BFRE UK model prior to the addition of any style factors. The top four share a common theme, namely, size. To decide how to aggregate these four candidate substyles to form a candidate size factor, we run additional univariate regressions on various combinations of these four candidate substyles as in Equation (4). This analysis suggests that the substyles Sales, Market Cap, Broker Coverage and Total Assets should all be aggregated together to form the size factor. We then add this style

factor into our first step regression in Equation (3) and re-run the two-step regression:

$$r = X_{\text{Mkt}} f_{\text{Mkt}} + \sum_{k \in \text{CCty}} X_{\text{CCty},k} f_{\text{CCty},k} + \sum_{j \in \text{CInd}} X_{\text{CInd},j} f_{\text{CInd},j} + X_{\text{size}} f_{\text{size}} + \eta \tag{5}$$

$$\eta = X_{\text{subSty},i} f_{\text{subSty},i} + \tilde{\eta}_i \tag{6}$$

The ranking of substyles after this regression is shown in Figure 5, which suggests Momentum is the next style factor to add to the model. We follow a similar procedure as in the previous stage to form the next style factor and incorporate this into the first-step regression in Equation (5). The same process is repeated until the largest proportion of t-statistics from the second step univariate regression is no larger than 10% - 15%. Throughout the whole selection process, qualitative judgement and statistical analysis are combined to identify substyle factors that explain asset returns and how they should be grouped together into a style factor.

Figure 4: proportion of significant t-statistics from the first stage regression

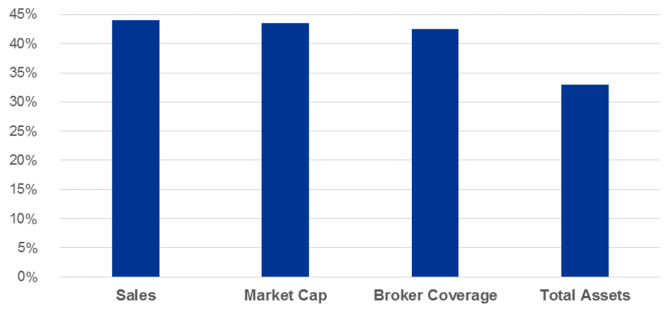
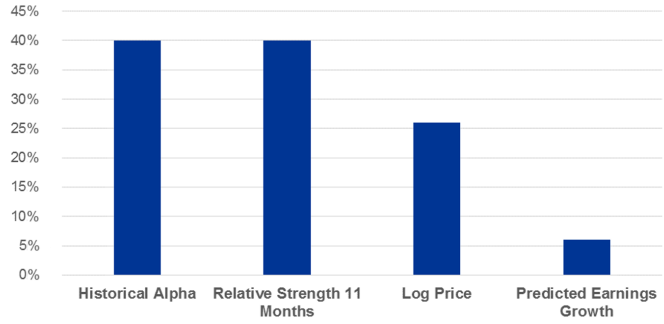


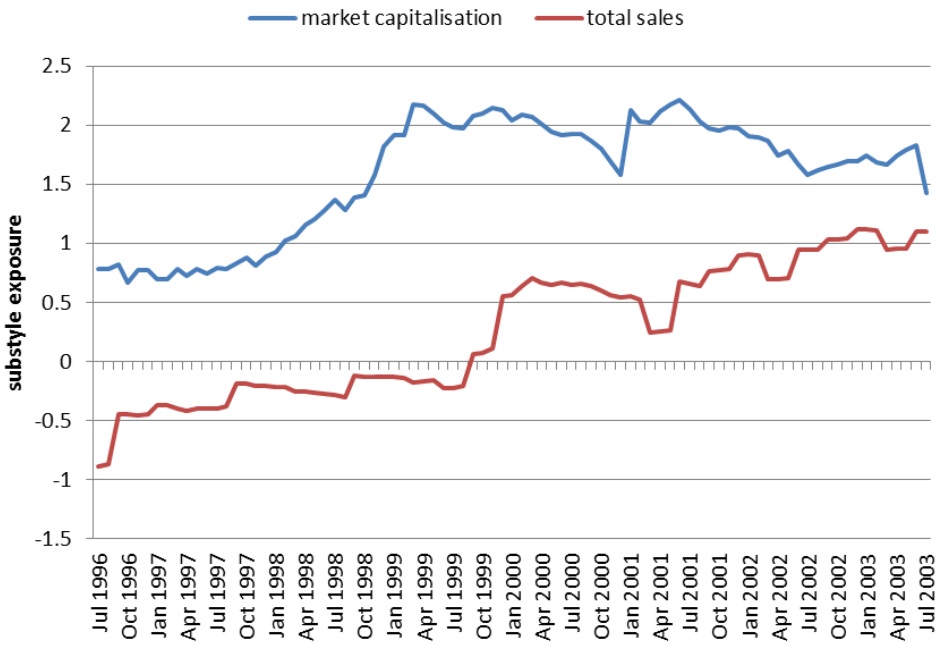
Figure 5: proportion of significant t-statistics from the second stage regression



Size The size factor captures the increased risks associated with investing in small-cap companies as compared with large-cap companies. Large companies are typically more immune to difficult economic conditions and general downturns in the business cycle, as compared with smaller companies. In this sense, similar sized companies behave in a similar fashion at different points of the business cycle, and so size has significant explanatory power in explaining asset return commonality.

Size is traditionally defined using market capitalisation; however, this only offers a single, and often incomplete, measure of the size of a company. In contrast, the BFRE definition uses different formulations of size based on balance sheet information, market data and analyst estimate data leading to a more robust definition. As an example, consider the size exposure of AOL Inc. in the North America model before and after the dotcom bubble. Figure 6 shows a significant disparity between two size-based substyles, market capitalisation and total sales. Market capitalisation is significantly positive at the beginning of the period, and increases

Figure 6: substyle exposures of AOL Inc., Jul 1996 – Jul 2003



substantially during 1998-9 following strong price performance. A measure of size based on market capitalisation alone would confer large-size status on AOL, and reduce its risk forecast accordingly. However, this fails to capture the underlying fundamentals, such as total sales, which describe a significantly smaller, and riskier, company relative to the market.

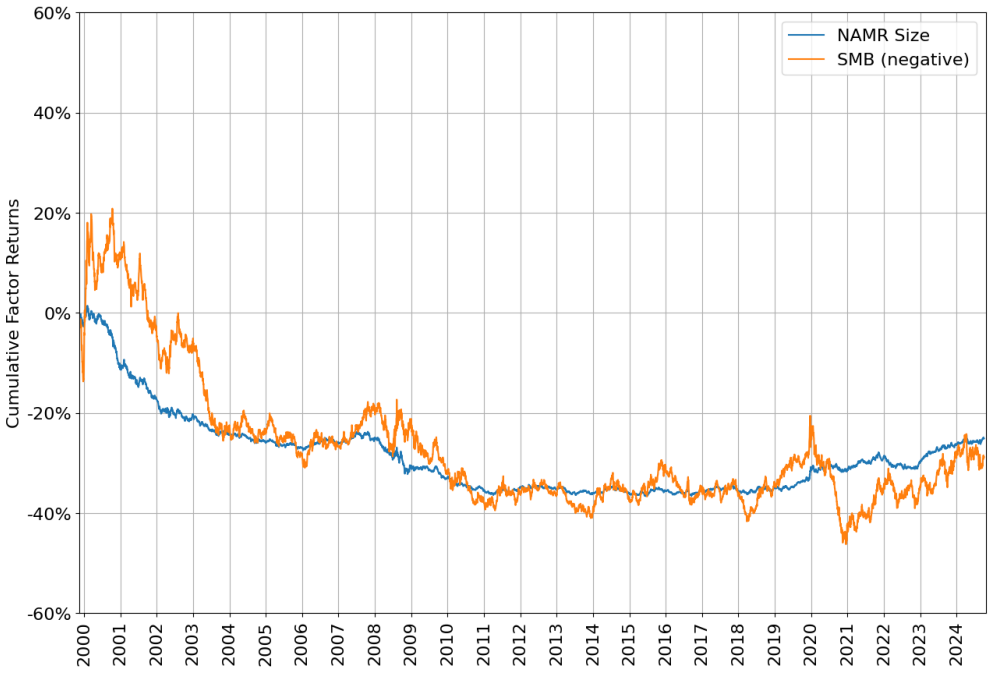
Size is also an important factor within the quantitative investment community since small-cap stocks have historically earned a risk premium relative to their large-cap peers (see Banz [6]). The under performance of large-caps relative to small-caps can be seen in the negative cumulative returns to the size factor in the NAMR model in figure 7 since 2000. This premium has decayed since 2010 and cumulative return trends flattened. Conceptually, this series represents the return to a portfolio of securities long large-cap and short small-cap stocks, after neutralising other style tilts, and industry/country effects in the model. Figure 7 also shows the cumulative returns to the Fama-French SMB ('small minus big') factor[12] for comparison. This series has been negated to align with the convention used in the BFRE size factor. Although not expected to match perfectly due to different definitions and construction, both series are closely related and have a correlation of 0.63 over the entire history.

Volatility The BFRE volatility factor is based on various measures of historical return dispersion, and historical beta. Historical beta is calculated via a univariate regression of asset returns against a capitalisation-weighted market index. The regression uses 5 years of weekly data and is exponentially weighted to place more emphasis on return behaviour in the previous 52 weeks.

Volatility is very highly correlated with the market (see Table 2) and is empirically proven to have significant explanatory power in explaining asset return commonality. Figure 8 shows the proportion of significant t-statistics for all style factors in the NAMR model[13]. This is a standard measure of factor efficacy and is one of metrics used to select which style factors are eligible for inclusion in BFRE. A value in excess of 10% indicates a statistically significant style effect, and would be considered for inclusion in the models. The proportion of significant t-statistics for Volatility is considerably higher than this threshold and larger than the other style

[12]Source: http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html
[13]t-statistics are based on daily cross-sectional regressions

Figure 7: cumulative returns to NAMR size and Fama-French SMB, Feb 2000 – Dec 2024



factors, and incidentally most industry and country factors, which shows the relative strength of the Volatility factor in explaining cross-sectional asset returns.

Large Volatility exposures tend to be associated with small-cap stocks through time. The correlation between Volatility and Size exposures in cross-section is -0.39 as shown in figure 3. This negative relationship is fairly stable over time. Consequently, portfolios which are positively exposed to size are commonly negatively exposed to Volatility, and vice versa.

Volatility factor exposures can also vary considerably across different sectors. Figure 9 shows the average Volatility exposure of securities in each sector. The IT sector, and to a lesser extent the Telecoms sector, exhibit the highest levels of Volatility exposure leading up to, and during, the dotcom crisis. Financials have relatively lower Volatility exposures in comparison, but this increases significantly during the financial crisis in 2007-8, at which point the average exposure changes sign to indicate above average levels of volatility in this sector. The Energy sector has significantly higher volatility during the COVID-19 pandemic in 2020 when oil prices plunged as well as 2022 when the geopolitical conflict led to supply chain shocks and soaring oil prices. Other sectors are relatively more stable, such as Utilities and Consumer Staples, which have consistently lower Volatility exposures over the entire period.

Mid-Cap & Small-Cap Research into the statistical significance of different market capitalisation deciles of the estimation universe reveals that there is additional asset return commonality present in the mid-cap and small-cap segments that is not fully captured by the existing size factor. Most BFRE models therefore also include a Small-Cap or Mid-Cap factor to incorporate this source of commonality.

Figure 10 shows the proportion of significant t-statistics for different market capitalisation deciles of the EMEA estimation universe, before and after the inclusion of a Small-Cap factor in the EMEA model[14]. For each decile, a set of model residuals were regressed against 0/1 dummy variables indicating decile membership. Prior to the addition of a Small-Cap factor, there is significant explanatory power in the lower deciles, 9 and 10, in excess of the 10% threshold used to determine whether styles are eligible for inclusion in the

[14]t-statistics are based on monthly cross-sectional regressions

Figure 8: proportion of significant t-statistics for NAMR style factors, Feb 2000 – Dec 2024

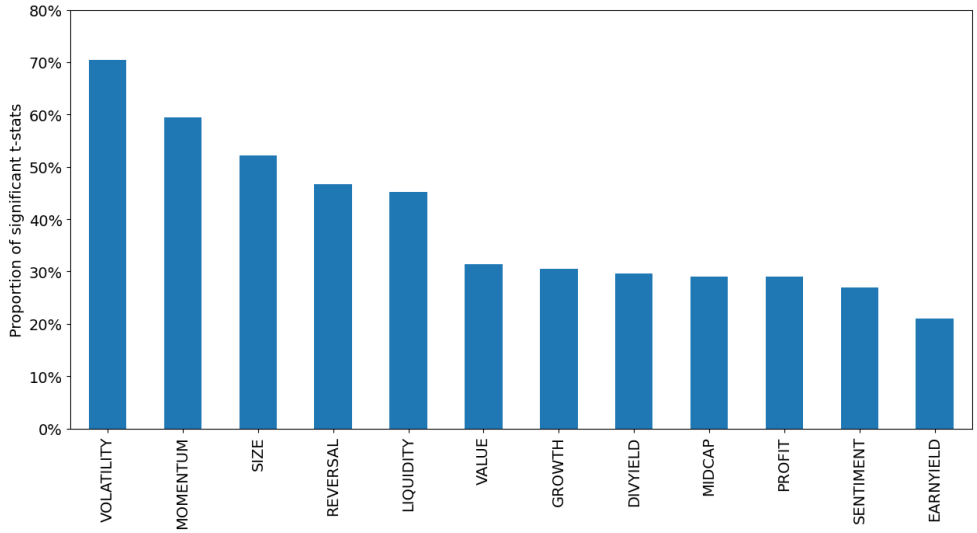


Figure 9: average NAMR Volatility factor exposures by sector, Feb 2000 – Dec 2024

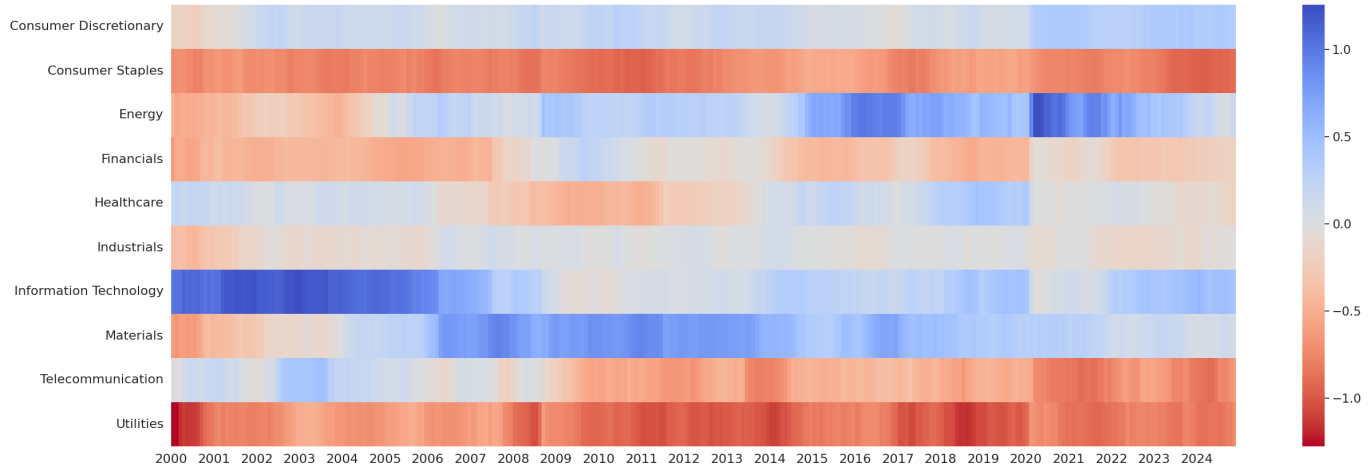
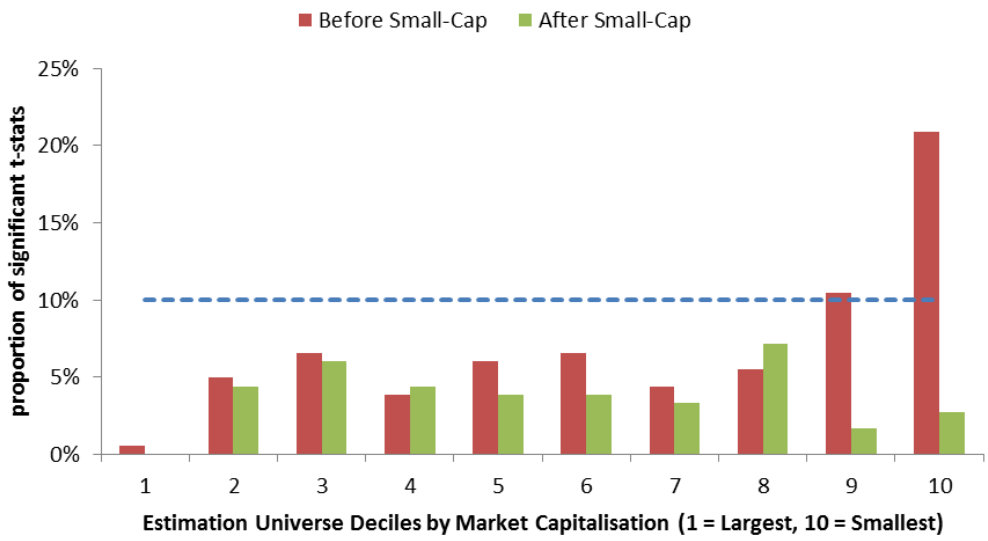


Figure 10: proportion of significant t-statistics for EMEA universe deciles, Mar 1996 – Dec 2010



model. Following the addition of a Small-Cap factor, the proportion of t-statistics for these deciles is no longer statistically significant.

The Small-Cap and Mid-Cap factors are defined as smoothed versions of dummy variables indicating membership to particular deciles of the model estimation universe. A smoothing function is used in preference to 0/1 indicators to mitigate instability in exposures for assets on the decile boundaries. The mid-cap factor primarily represents decile 6 and 7, whilst the Small-Cap factor represents assets in deciles 9 and 10. See Equations (49) and (50) for more details on the exact functional form.

Small-Cap was found to be significant in the WRLD, EMEA, APXJ, AUST, UKIN, USAM and EMKT models, whilst Mid-Cap was found to be significant in the NAMR model.

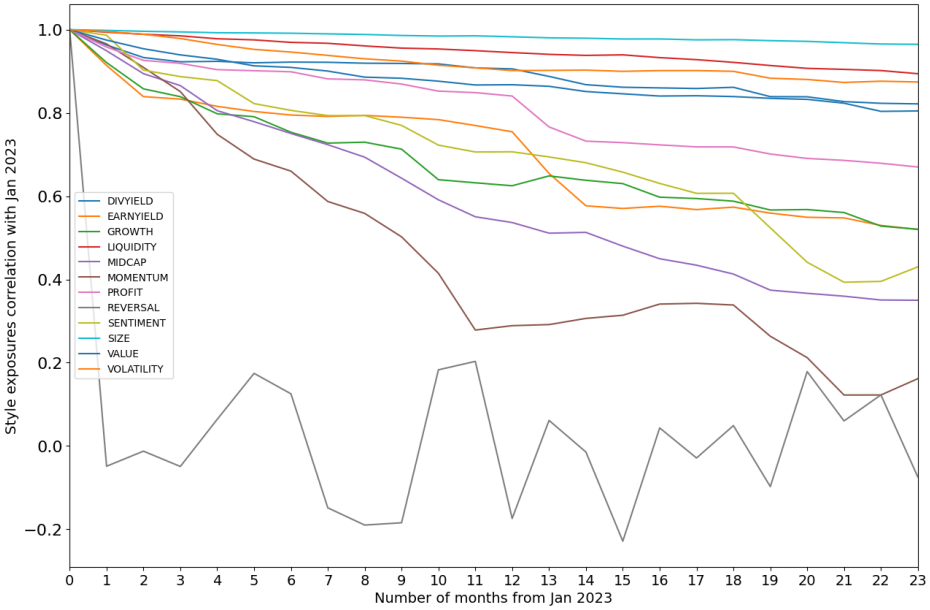
Reversal & Momentum The Reversal and Momentum factors are based on anomalies identified in stock market research. Both factors have significant power in explaining the cross-sectional equity returns, and by extension, asset return covariance and beta. Additionally, both are also associated with significant returns in excess of the market which cannot be explained using other common factors.

The Reversal factor, documented in Jegadeesh [7] and Jacobs and Levy [8], captures a near-term reversal in strong stock price performance measured over a one-month period. There are a variety of explanations offered in the academic literature for the source of this phenomenon. A popular, behavioural interpretation links this effect to market investors overreacting to stock information in the near-term, e.g. see Subrahmanyam [9].

The Momentum factor is a related effect, first documented in Jegadeesh and Titman [10], and captures the tendency of previous stock winners to continue rising, and for previous stock losers to continue losing. The momentum effect has been observed over various horizons, typically ranging from three months to 12 months. The BFRE definition focuses on stock returns over the previous 11 months with a one-month lag to exclude the reversal effect. Similar to reversal, there is a behavioural explanation for this anomaly attributing it to market investors systematically underreacting to newly available company information, and failing to adjust their expectations in the medium-term.

In contrast to most style factors in BFRE, the Reversal and Momentum factor exposures of a security can vary considerably through time. Figure 11 shows the cross-sectional exposure correlations of each style in January 2020 against its future values for every month in the next two years, i.e. the correlation of style exposures in Jan 2020 with the next month, next 2 months, and so on. Correlations close to one indicate a high degree of persistence in asset style exposures through time. Reversal and Momentum clearly exhibit the least

Figure 11: style exposure autocorrelations in the NAMR model, Jan 2023 – Dec 2024



persistence relative to other styles, which reflects the highly time-varying nature of these factor exposures. Due to the fast changing nature of the Reversal factor, it is only included in BFRE 1.0 and BFRE 2.0 Short Horizon models, but not featured in the BFRE 2.0 Long Horizon models.

It is well-documented in the above academic references that a momentum-based portfolio strategy is rewarded with significant positive returns through time, and likewise, a reversal strategy with negative returns. Table 2 shows that Momentum and Reversal have larger Sharpe ratios relative to the Market factor. The Table also indicates that Momentum and Reversal returns are associated with a significant first-order autocorrelation. This can be seen clearly in a plot of the cumulative returns to the NAMR Momentum factor, as shown in Figure 12. The stable upward trend in Figure 12 is first significantly interrupted during the Quant crisis in end of 2007 where the series begins to flat-line prior to suffering a significant drawdown during the financial crisis period. Thereafter, the series continues on its characteristic upward trajectory until another interruption in 2015. The series has mixed performance up to the 2020 COVID-19 period, where it suffered another major drawdown.

Some multi-factor risk models combine Reversal and Momentum into a single factor; however, the BFRE models separate these into different factors. This decision allows model users to isolate these effects in risk decomposition and factor attribution analyses. This separation also avoids biases in portfolio optimisations for quantitative equity portfolios that invest in assets heavily tilted towards these factors (see Lee and Stefek [11]).

Liquidity The Liquidity factor captures the differential return and risk of more liquid stocks relative to less liquid stocks controlling for industry, country and other style effects. Liquidity is not directly observable and can relate to a number of different stock characteristics (see Sarr and Lybek [12] or Connor et al. [13]). Liquid stocks typically entail low transaction costs (e.g. low bid-ask spreads); trade frequently at large volumes with minimal price impact; and can be converted into cash quickly. Given there is no single, widely-accepted metric that captures all of these dimensions, the BFRE Liquidity factor uses multiple substyles in its definition.

The definition of liquidity varies depending on the region or country of interest, and can use a combination of volume-based measures, and price impact measures. Volume-based measures, such as share turnover or active trade days, distinguish liquid stocks by the magnitude and frequency of trades and in doing so capture market breadth (Bekaert et al. [14]). The price impact measures, such as the Amihud ratio, the return-to-

Figure 12: cumulative performance of Momentum in the NAMR model, Feb 2000 – Dec 2024

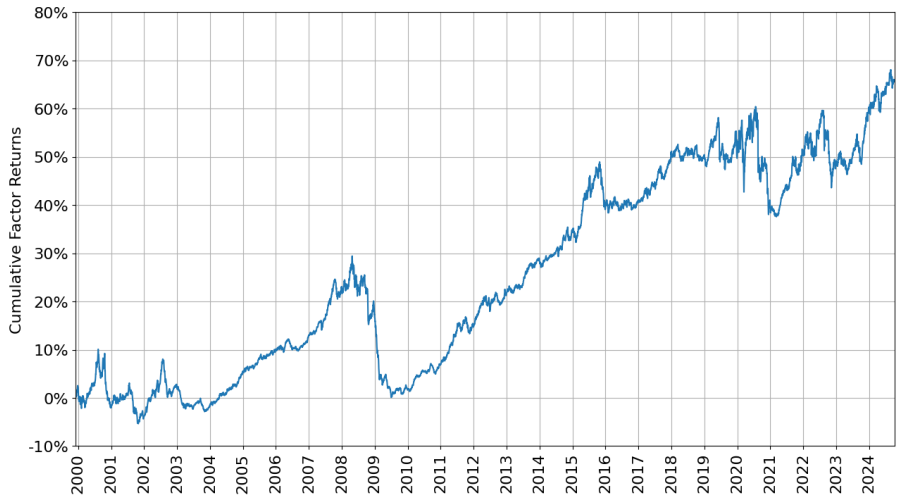
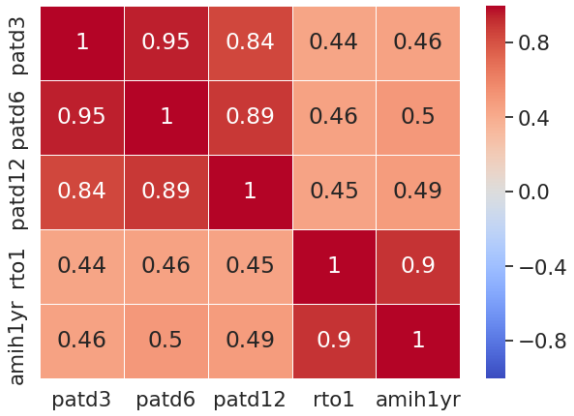


Figure 13: substyle exposure correlations in the NAMR model, Dec 2024

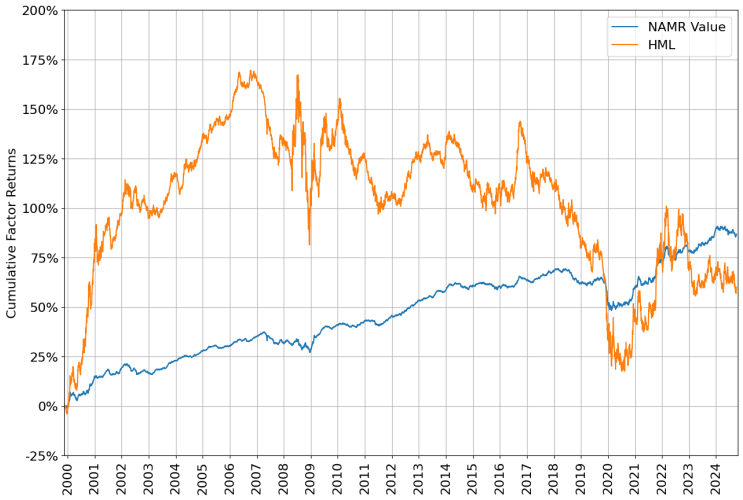


turnover ratio (RTO) and price pressure are taken from the academic literature (see Amihud [15], Florackis et al. [16] and Bekaert et al. [14]). Amihud ratio and RTO substyles relate price movement with the volume of transactions and identify illiquid securities as those which exhibit larger price moves for the same level of volume. Price pressure relates price impact with lengthy periods of consecutive non-trading days and identifies assets prone to a subsequent return 'catch-up' effect. The literature also documents liquidity measures based on higher frequency, intraday data, however, BFRE only uses low-frequency data as these are more readily available for the different markets covered by BFRE. Additionally, Goyenko et al. [17] show the Amihud ratio has a significant positive correlation with some of these alternative measures.

Although these measures capture different aspects of liquidity, they are nevertheless strongly related. Figure 13 shows the correlation between the various substyles used to define Liquidity in the NAMR model. There is significant positive correlation amongst, as well as between, the various volume-based and price impact measures of liquidity.

Liquid securities generally have a higher correlation to the market – Table 2 shows that the correlation between the Liquidity and Market factors in the NAMR model is 0.65. Liquid stocks tend to react more quickly to general economic news flow and changes in market sentiment reflecting the increased levels of trading activity and investor focus on this segment of the market. Liquidity is also highly correlated to the size factor, as shown in figure 3, i.e. the securities of large companies tend to be more liquid than those of smaller companies. This

Figure 14: cumulative performance of Value in the NAMR model, Feb 2000 – Dec 2024



relationship is stable over the entire research history.

Value & Growth Value is a highly pervasive risk factor defined using standard valuation ratios: book-to-price, sales-to-price and cash flow-to-price. Elementary pricing theory suggests that stocks with high fundamental value relative to market value (i.e. 'cheap' stocks) exhibit differential interest rate risk and growth risk compared to stocks with lower valuation ratios (i.e. 'expensive' stocks). As market expectations of interest rates and economic growth vary, so too does the relative pricing of these two groups of stocks. Hence value factors are useful for explaining sources of asset return commonality.

Value is also associated with a significant positive risk-adjusted return over time – referred to as the value anomaly in asset pricing theory. This finding is robust with respect to different markets and time periods. Table 2 shows the Value factor has a Sharpe ratio of 1.08 in the NAMR model and exhibits a significant first-order autocorrelation. Figure 14 shows the cumulative returns to the Value factor in the NAMR model. The strong upward trend reflects the high degree of persistence in the Factor Returns up to 2018 before flattening in 2019 and suffering a drawdown in 2020. This figure also includes the cumulative return series to the Fama-French HML ('high minus low') factor for comparison. The general profile of HML closely resembles that of the NAMR Value factor. Both have a significant positive correlation of 0.52 over the entire period.

The Growth factor can be used as a proxy for the economic and business cycle risks of a company. Companies that grow more quickly have differential exposure to interest rates and economic growth, and will carry different risks at different points in the business cycle. The Growth factor therefore helps to explain asset return commonality.

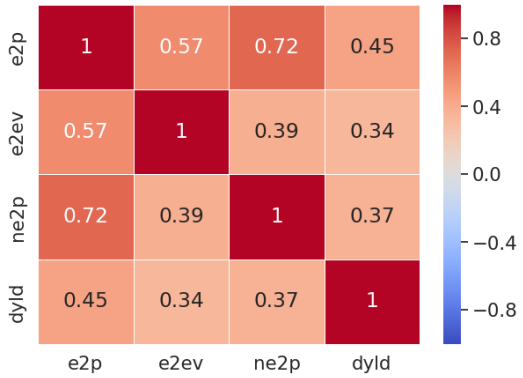
The BFRE definition varies by region and country, but typically includes some measures of historical asset growth and sales growth. In addition, the EMEA and NAMR models also employ a forward-looking measure of sales growth based on analyst estimates.

Earnings Yield, Dividend Yield & Yield Earnings and dividends act as proxies for value and can be used to identify cheap and expensive stocks in the market. As earnings are highly cyclical and dividends are relatively stable, both can provide a useful gauge of the relative valuation of a company at different points of the business cycle.

The Earnings Yield factor is based on different substyles using the latest reported earnings and captures asset return commonality between companies with similar earnings-to-price ratios. The Dividend Yield factor likewise captures commonality between companies with similar dividends-per-share.

Earnings Yield and Dividend Yield appear separately in the BFRE models, with the exception of the EMEA

Figure 15: substyle exposure correlations in the EMEA model, Dec 2024



model, where they were found to be highly correlated over the research history and so were combined into a single factor, referred to as Yield. This is evidenced in figure 15, which shows the correlations between the various substyles that define the Yield factor in the EMEA model. The BRAZ and LATC models show Yield instead of Earnings Yield in the front end tools: Explore and Portfolio Risk Tools (PRT). Exact style definitions are shown in Table 8.2.

Profitability The Profitability factor is based on several standard profitability ratios which measure the magnitude of company earnings relative to expenses, such as return-on-equity (ROE), return-on-assets (ROA) and interest coverage. Conventionally, these ratios measure the quality, sustainability and stability of a company. Companies which rank highly are considered to be more safe and stable relative to those that have lower scores.

Profitability can be positively correlated to Value, Earnings Yield and Dividend Yield. This can be understood by considering typical examples of low value companies, such as early stage technology companies, which have low margins and low ROE, and so would score poorly on measures of profitability. Profitability however differs from value factors in that it is computed purely on the basis of fundamental data. No market data is used to compute the Profitability exposure of a company. It therefore adds another risk dimension to the factor structure of the models.

Leverage The Leverage factor captures the relationship between stock returns and the degree to which a company's balance sheet is leveraged. At certain times, particularly during periods of market distress and heightened risk aversion, high leverage companies experience increased commonality in returns.

The BFRE definition uses the following standard measures of balance sheet leverage: debt-to-assets ratio, book leverage, market leverage and the proportion of cash relative to total assets.

Quality The Quality factor is taken from the academic literature relating increased levels of share issuance with subsequent negative performance in stock returns. Empirical evidence for this effect is documented for US returns in Daniel and Titman [18], and for international markets in McLean et al. [19].

Several quality substyles, including those measuring earnings variability, were tested in each model (see Table 6); however, only equity dilution was found to have significant cross-sectional explanatory power in the Asia Pacific ex Japan (APXJ) model.

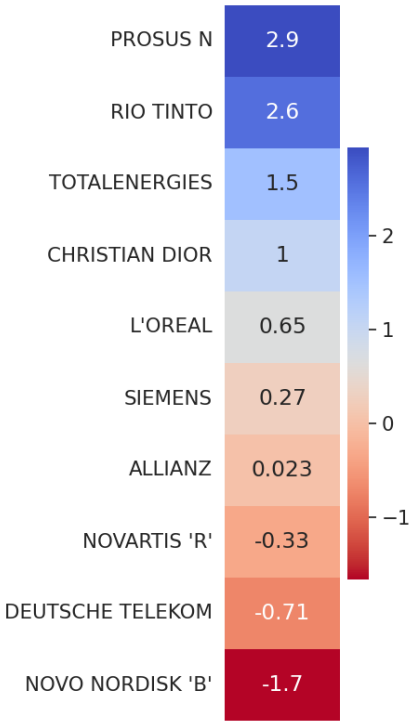
Emerging Market For a large number of developed market companies, revenue growth is increasingly dependent on operating activities in emerging markets. The Emerging Market factor captures this dependency through a regression analysis of the asset returns on emerging market performance.

The Emerging Market sensitivity is defined as the beta of residual asset returns against the outperformance of the iShares MSCI Emerging Market ETF relative to the S&P 500. The regression uses 5 years of weekly data and is exponentially weighted to place more emphasis on return behaviour in the previous 52 weeks. Asset

residual returns are formed from a univariate regression of asset returns against a capitalisation-weighted market index.

To aid with interpretation, it is useful to consider which stocks are highly exposed to the Emerging Market factor. Figure 16 shows the Emerging Market factor exposures for a number of headline companies in the EMEA model. Companies with significant operating activities in emerging markets, e.g. Rio Tinto and/or those with a high proportion of sales/revenues from emerging markets, e.g. Christian Dior, and/or those investment groups with significant investments in companies in emerging markets, e.g. Prosus, are highly exposed to the Emerging Market factor. Furthermore, companies such as Deliveroo that only serve developed markets bear very little exposure to this factor.

Figure 16: Emerging Market factor exposures in the EMEA model, Dec 2024



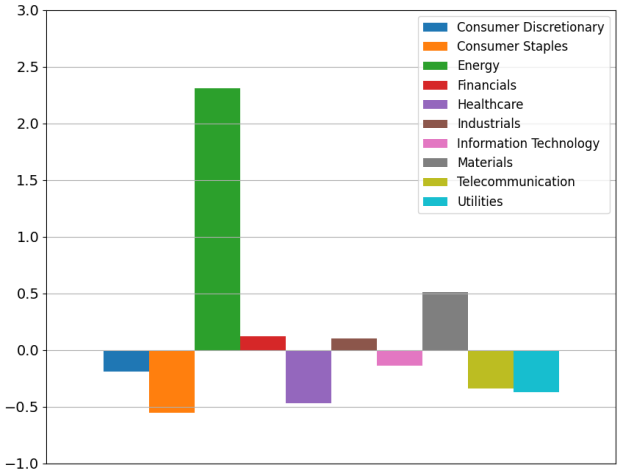
Developed Market Analogous to the Emerging Market factor definition. The Developed Market factor identifies Emerging Market companies whose revenue growth is increasingly dependent on operating activities in developed markets. The Developed Market factor captures this dependency through a regression analysis of the asset returns on developed market performance.

Oil Stock markets can be significantly impacted by the rise and increasing volatility of oil prices. Oil is a significant macro-economic factor and is closely related to cycles in business activity but can also be impacted by heightened levels of risk aversion in financial markets, e.g. as a result of wars, or even the fear of wars. The BFRE Oil factor captures return commonality due to assets that are more prone to benefit or suffer from these changing oil price dynamics.

The Oil factor exposure is defined as the beta of residual asset returns against the oil price. Similar to the emerging market factor exposure, the Oil factor exposure uses a 5-year history of weekly asset residual returns in excess of the market. Figure 17 shows the average Oil factor exposure of securities grouped by sector as of December 2022. As expected, securities in the energy and materials sectors show the largest positive exposure

to oil price fluctuations over the entire history. The remaining sector exposures are more muted with average values close to zero.

Figure 17: average Oil factor exposure for securities grouped by sector, Dec 2024



Sentiment The recent past has been characterised by a rapid succession of 'risk-on/risk-off' trading. Some securities are more prone to follow these trends whereas others are less sensitive to this market sentiment. A regression analysis between asset returns and changes in the level of the VIX index (as a first approximation to the 'risk-on/risk-off' paradigm) captures this sensitivity. The details of the regression are similar to the other macro-economic factors in the model. See Equation (51) for more details.

3.2 Model Estimation

In general a multi-factor model decomposes asset returns as follows:

$$r = Xf + u \tag{7}$$

where

r	: asset returns (in excess of the local risk-free rate)
X	: set of factor exposures based on asset characteristics
f	: return to each factor
u	: asset specific return

Xf is termed the common factor return and u is the asset specific, or idiosyncratic, return. In a well-specified model, the prescribed factors will capture all sources of commonality in asset returns leaving only return associated with stock-specific events in the idiosyncratic component.

There are a variety of methods for estimating linear factor models of this form. The approach taken for BFRE is similar to that used in the fundamental factor risk model literature[15]. The estimation phase involves a series of cross-sectional regressions of asset returns against asset factor exposures, which provides estimates of Factor Returns and asset specific returns. These regressions are performed daily for country and regional models and weekly for the World model. The resulting Factor Returns are used to estimate a factor covariance matrix, and the asset specific returns are used to estimate asset specific risk forecasts. Portfolio risk can then be calculated using portfolio holdings information, i.e. assets weights, portfolio-level factor exposures (aggregated from the asset level), a factor covariance matrix and asset specific risk forecasts. The last three components are represented in the 'risk' version of the above stylised Equation as:

$$\Sigma = XFX^{\top} + \Delta \tag{8}$$

where

Σ	: asset covariance matrix
F	: factor covariance matrix
Δ	: specific risk matrix (a diagonal matrix of asset specific risk forecasts)

A discussion of the factor covariance matrix and asset specific risk methodology is deferred until later; instead the current section focuses on the details of the model estimation.

Estimation Universe

The estimation universe comprises the set of assets that are used during the model estimation process to infer the Factor Returns and asset specific returns. It is constructed on a country-by-country basis every month, and is defined to include only representative assets from each market. In particular, illiquid assets which may have stale prices that do not reflect their true market values are excluded, as are 'tiny' companies whose economic relevance is questionable.

Estimation universe assets are screened using information on security type, liquidity and market capitalisation. The methodology used to construct the estimation universe is similar to that used by index providers, and so there is a significant amount of overlap between the assets in the BFRE estimation universes and those found in comparable third-party indexes covering the same region.

[15] See Rudd and Clasing [20], Grinold and Kahn [21] and Connor et al. [13] for more details

Factor Return Estimation

The model estimation is performed in two passes. The first-pass uses the most important assets in a region (or country) to estimate the most important factors. It excludes assets in markets with lower development status, where data quality and coverage is considerably poorer. Instead, these assets are relegated to the second-pass of the model estimation, and therefore have no impact on the Factor Returns estimated in the first-pass.

Assets are weighted in the regression using square-root of market capitalisation. This provides a good compromise between using a weighting scheme that places equal weight on all assets, and one that is weighted by market capitalisation. The former can place too much emphasis on smaller assets in the regression and can lead to a poor fit for mega-cap and large-cap securities, many of which feature prominently in portfolios and benchmarks. Likewise, using (only) market capitalisation may lead to better risk forecasts for mega-caps but would come at the expense of poorer forecasts for mid-cap and small-cap securities. More technically, square-root of market capitalisation adjusts for heteroskedasticity based on the observation that higher residual (specific) risk is typically correlated with smaller market capitalisation assets[16].

The first-pass of the factor return estimation is used to estimate Market, Style, Core Industry and Core Country factors. Core Industry factors include all industries except the Multi-Sector Holding industry (if applicable) as these assets are typically nothing but amalgams of other assets that may already be in the estimation universe. Core Countries are those classified as Developed or Advanced Emerging. These countries are considered to be more representative in the region and so are given higher prominence in the model estimation as compared with Secondary Emerging, Frontier and Watch markets, collectively referred to as Extended Countries. The Extended Country Factor Returns are estimated in the second-pass of the model estimation.

In the BFRE 2.0 models, Short Horizon models and Long Horizon models have the same factor return estimation methodology with one difference in factor structure: the Reversal factor is only included in the Short Horizon models but removed from the Long Horizon models.

The first-pass of the factor return estimation can be represented as follows:

$$r = X_{\text{Mkt}} f_{\text{Mkt}} + \sum_{i \in \text{Sty}} X_{\text{Sty},i} f_{\text{Sty},i} + \sum_{j \in \text{CInd}} X_{\text{CInd},j} f_{\text{CInd},j} + \sum_{k \in \text{CCty}} X_{\text{CCty},k} f_{\text{CCty},k} + u \tag{9}$$

where

r	: local asset returns in excess of the risk-free rate
$X_{\text{Mkt}} f_{\text{Mkt}}$	: Market exposures and factor returns
$X_{\text{Sty},j} f_{\text{Sty},j}$	: style exposures and factor returns
$X_{\text{CInd},j} f_{\text{CInd},j}$	: core industry exposures and factor returns
$X_{\text{CCty},k} f_{\text{CCty},k}$	: core country exposures and factor returns
u	: residual returns from the core factor estimation

This specification is not uniquely identified as there are an infinite number of possible solutions. The reason for this lies in the fact that, for each asset, there exist three intercept terms – the Market factor, an industry factor, and a country factor. Every asset has unit exposure to these three factors. This can be resolved by adding two linear restrictions on the definition of the Factor Returns, which reduces the intercept terms from three to one. This identification simply represents a rotation of the Factor Returns. It does not impact the efficacy of the risk model.

The restrictions force the average[17] industry and country returns to be zero, and take the following form:

$$\sum_j \omega_{\text{CInd},j} f_{\text{CInd},j} = 0 \qquad \sum_j \omega_{\text{CCty},k} f_{\text{CCty},k} = 0 \tag{10}$$

[16] Other regression weighting schemes exist which explicitly use the reciprocal of each asset's residual (specific) variance, however, in practice these provide similar results to using square-root of market capitalisation weights

[17] The average return is a square-root of market capitalisation weighted return

where

$\omega_{\text{CInd},j}$	: the average (square-root of capitalisation) weight for core industries
$\omega_{\text{CCty},k}$	: the average (square-root of capitalisation) weight for core countries
$f_{\text{CInd},j}$	: core industry factor returns
$f_{\text{CCty},k}$	: core country factor returns

As a result of these restrictions, the industry and country factors are net of the Market factor return, which impacts their interpretation. To illustrate, if asset returns across EMEA are mostly positive on a given day and UK assets are also up but by less than the average return over the region, then the UK factor return will be negative.

In the case of single-country models, such as Japan, there are no country factors since the Market factor is defined on a single country. In this instance only one linear restriction is required – that for the industry factors.

Once the first-pass has been executed, a set of residuals is computed for all assets used in the second-pass to estimate the Factor Returns on all Extended Country factors and Extended Industry factors. This can be represented as follows:

$$u = \sum_{j \in \text{EInd}} X_{\text{EInd},j} f_{\text{EInd},j} + \sum_{k \in \text{ECty}} X_{\text{ECty},k} f_{\text{ECty},k} + \epsilon \tag{11}$$

where

u	: residual returns from the core factor estimation
$X_{\text{EInd}} f_{\text{EInd}}$	: extended industry exposures and factor returns
$X_{\text{ECty}} f_{\text{ECty}}$	: extended country exposures and factor returns
ϵ	: residual returns from the extended factor estimation

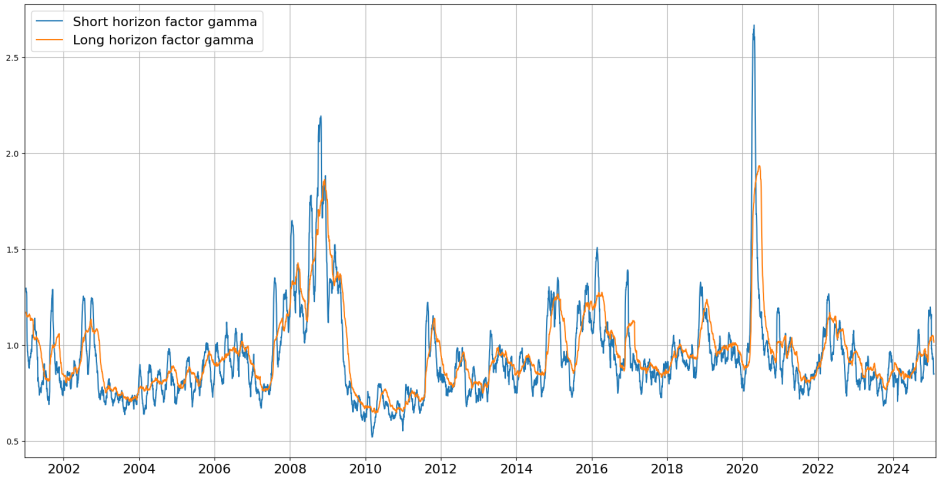
Factor Covariance Matrix

The factor covariance methodology is chosen in accordance with best practice based on theory, experience and stylised facts relating to financial time-series. Observations are weighted using an exponential decay (EWMA) that places more emphasis on recent Factor Returns to give more responsive risk forecasts. The responsiveness is controlled through the half-life of the exponential decay. The value of the half-life is primarily driven by the target horizon, but can also reflect issues relating to forecast accuracy and stability. Ultimately, forecasts of factor volatilities and factor correlations should be responsive to changes in the market environment whilst not being unduly noisy so as to render them unstable and unusable.

See Table 3 for the default factor covariance matrix calculation parameters. However, to accommodate different horizons and use-cases, Aladdin provides the flexibility to re-estimate the factor covariance matrix. This can be done using Portfolio Risk Tools (PRT) and Explore. PRT and Explore allow users to change the lookback and half-life of the exponential decay in the factor return series, and apply different adjustments to factor volatilities and factor correlations. In contrast with the factor covariance matrix being able to be re-estimated on the fly in PRT and Explore, the asset specific risk forecasts are fixed and cannot be changed through the front end for the time being. The default factor covariance matrix is set-up in such a way that is guaranteed to be positive semi-definite. Model users are advised that the above mentioned factor covariance matrix treatments are only available for specific settings in PRT and Explore. This is similarly the case for the BFRE2.0 covariance matrix enhancements as described below.

BFRE 2.0 enhancement While the BFRE 1.0 factor covariance estimation is based on EWMA approach with the strengths of simplicity and robustness, it has the limitation that factor volatility forecast is informed solely

Figure 18: factor risk gamma γ scaling in NAMR 2.0 Short Horizon and Long Horizon model



with reference to prior Factor Returns and does not capture clustering behaviour in response to sudden shifts in market conditions. The BFRE 2.0 models leverage the 1.0 EWMA forecasts with a fixed set of parameters presented in Table 3, with two additional enhancements to produce more responsive forecasts while maintaining stability.

Table 3: BFRE factor covariance parameters

Model frequency	Variance Half-life	Correlation Half-life	Observations
Daily, 1.0 model	250 days	500 days	expanding window
Daily, 2.0 Short Horizon	250 days	500 days	2000 days
Daily, 2.0 Long Horizon	250 days	500 days	2000 days
Weekly (WRLD and EMKT), 1.0 model	52 weeks	156 weeks	expanding window
Weekly (WRLD and EMKT), 2.0 Short Horizon	52 weeks	156 weeks	416 weeks
Weekly (WRLD and EMKT), 2.0 Long Horizon	52 weeks	156 weeks	416 weeks

This section describes the two-step enhancement in more detail. Starting from the EWMA forecasts from BFRE 1.0 models described in the previous section, a multiplicative scaling factor, gamma γ, is applied to the 1.0 factor covariance matrix. The gamma γ scaling is model-specific and estimated on a daily basis to dynamically adapt the covariance matrix to shifts in recent market conditions. While γ has a long run average value of 1 indicating neutral impact in times of stable volatility, it comes into play when volatility conditions change sharply. Figure 18 shows the estimates of γ rapidly rises above 2 when market volatility escalates, spiking during the 2008-2009 Great Financial Crisis (GFC) and to an even greater level in the 2020 COVID-19 outbreak. The gamma scaling descends rapidly below 1 after the peaks as the volatility regime reverts back to longer run levels. Short and Long Horizon risk forecasts require different degrees of adaptation: gamma γ in Long Horizon models adjusts less rapidly and aggressively and the evolution over time is more stable than the Short Horizon counterpart. The same gamma scaling is applied to all factors in the model. Details of the calculation of factor risk gamma scaling can be found in the Appendix 8.6.

Second, the EWMA forecasts and gamma γ adjusted forecasts are combined linearly with equal weights to give the final factor covariance matrix, as shown in Equation below. Research by Timmermann [22] suggests that simple linear combinations of models are likely to outperform standalone models in unstable environments. Testing shows the combination forecast yields both out-of-sample forecasting improvements and

stability gains - consistent with the suggestions from academic literature [22].

$$F = 0.5F^{EWMA} + 0.5F^{\gamma-adj} \tag{12}$$

where

F	: BFRE 2.0 factor covariance matrix
F^{EWMA}	: time series based EWMA factor covariance matrix
$F^{\gamma-adj}$	: dynamically γ-adjusted factor covariance matrix

Specific Covariance Matrix

The asset specific covariance matrix is made up of two components: a vector of asset specific risk forecasts and a sparsely populated unit diagonal matrix containing non-zero, off-diagonal specific return correlations.

Specific Risk Specific risk forecasts are primarily estimated using a time-series of specific returns. Specific returns are calculated using local asset excess returns, factor exposures and (estimated) Factor Returns, and are then cleansed using robust methods to down-weight the influence of outlying observations.

The specific risk model uses similar modelling techniques as the factor covariance matrix. Specific returns are weighted using an exponential decay, and the Newey-West [23] methodology is used to aggregate daily specific returns to form specific risk forecasts that account for serial correlations in the daily data for the daily models. For the weekly models the same methodology is used with weekly specific returns. The choice of decay parameters is likewise determined by balancing the conflicting demands of having accurate and responsive forecasts against having stable ones. Table 5 shows the exponential decays expressed as half-lives for the different frequency of model, the number of observations used in the calculation and the number of lags used in the Newey-West serial correlation adjustment.

Table 4: BFRE specific risk parameters

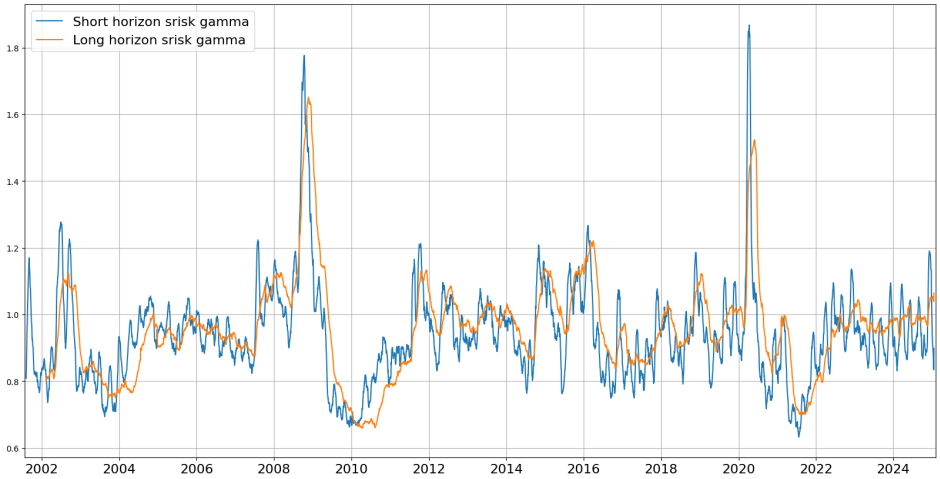
Model frequency	Half-life	Observations	Newey-West Lag
Daily, 1.0 model	125 days	375 days	10 days
Daily, 2.0 Short Horizon	125 days	375 days	1 day
Daily, 2.0 Long Horizon	250 days	500 days	1 day
Weekly (WRLD and EMKT), 1.0 model	26 weeks	104 weeks	2 weeks
Weekly (WRLD and EMKT), 2.0 Short Horizon	26 weeks	104 weeks	1 week
Weekly (WRLD and EMKT), 2.0 Long Horizon	52 weeks	208 weeks	1 week

The above approach is predicated on having a sufficient history of data. For assets with little or no history, e.g. IPOs, this is not the case. Therefore, the specific risk model includes a cross-sectional overlay which is used to infer missing specific risk forecasts based on (non-missing) forecasts for assets with a similar market capitalisation, in the same industry and country. The final specific risk forecast for every asset is then formed as a weighted sum of its time-series forecast (if it exists) and its cross-sectional forecast. The weighting function places more weight on the time-series forecast as more data becomes available. In the limit this weight is set to 1 so that every asset eventually has a specific risk forecast based on the time-series estimate alone.

BFRE 2.0 enhancement The 2.0 specific risk leverages a linear combination of a time series and structural approach.

$$\sigma_i = 0.5\sigma_i^{EWMA} + 0.5\sigma_i^{STR} \tag{13}$$

Figure 19: specific risk gamma γ scaling in NAMR 2.0 Short Horizon and Long Horizon model



where

σ_i : BFRE 2.0 specific risk of asset i

σ_i^{EWMA} : time series specific risk of asset i, based on exponentially weighted specific returns

σ_i^{STR} : dynamically γ-adjusted structural specific risk of asset i

Similar to the methodology of gamma adjustment for factor covariance matrix, the time series forecast from 1.0 specific risk described in the previous section is also adjusted by a gamma γ scaling to improve responsiveness of asset-specific risk to recent market conditions. While factor risk gamma γ scaling is estimated from recent realised factor volatilities, the specific risk gamma γ scaling is estimated from recent realised asset-specific volatilities on a daily basis. Details of the calculation of gamma scaling can be found in the Appendix 8.6. Figure 19 shows the estimates of specific risk gamma γ scaling are consistent with factor risk gamma γ scaling, spiking during the 2008-2009 Great Financial Crisis (GFC) and 2020 COVID-19 outbreak and quickly reverting to below 1 post crisis.

As the first step in the specific risk enhancement, the same specific risk gamma γ scaling is applied to all assets in the model coverage universe. This assumes all assets respond similarly to changes in common volatility conditions, which is unlikely to be true. Like asset returns, asset specific volatilities exhibit strong factor structure and common sources of variations. The degree to which γ scaling affects the specific risk forecast of one asset is a function of characteristics, defined by a subset of factor exposures. For Short Horizon models, Size, Volatility, Liquidity and Momentum are determinants of this sensitivity while for Long Horizon models, Size and Volatility are used to compute a security-level adaptation to current risk conditions. This adjustment produces the structural specific risk forecast σ_i^{STR}. Details of the structural forecast methodology can be found in the Appendix 8.7.

To illustrate the impact of BFRE 2.0 enhancements on factor risk and specific risk, Figure 20 and 21 show the contribution to analytical risk report in Aladdin, on date 27 March 2020, comparing the risk estimates of MSCI US index using the BFRE 1.0 NAMR model vs. the BFRE 2.0 Short Horizon NAMR model. The date is chosen to be around the start of the COVID-19 pandemic when the market volatility is spiking. The corresponding factor risk gamma scaling and specific risk gamma scaling are around 2 as shown in Figure 18 and 19, which meant the factor risk and specific risk are roughly doubled. This translates to the total portfolio risk estimated using the BFRE 2.0 NAMR Short Horizon model to be about twice that estimated using the corresponding 1.0 model.

Specific Return Correlation Linkages between related assets in the same company are not completely captured by the common factors in the model. In these instances, specific returns continue to capture both the idiosyncrasies of asset return and company return. As a result these specific returns may be positively correlated, and so ignoring this correlation would lead to under (or over) prediction of specific risk in a long-only (long-short) portfolio context.

Examples of such related assets include cross-listings of the same security; ADRs and their root assets; and different share classes, e.g. 'A' and 'B' shares in Sweden, and 'Registered' and 'Bearer' shares in Switzerland.

In all circumstances linkages can be captured through specific return correlations, either imposed via a constant override of 1, or where data availability permits, estimated using asset specific returns. Note that specific return correlations are only estimated between assets in the same company. Specific return correlations between assets in different companies are assumed to be zero in-line with standard modelling practice.

Structural & Empirical Models The BFRE models use two different methodologies to construct the asset specific covariance matrix. These differ only in their treatment of related listings in the same company. The first approach is referred to as the structural approach and assigns related listings identical specific risk forecasts (typically cloned from the primary listing) and forces the specific return correlation to be 1. In contrast, the second – empirical– approach estimates these quantities separately using data specific to each asset, and so results in different specific risk forecasts and specific return correlations. These approaches impact the following related asset groups: (a) cross-listings and (b) derived securities (ADRs, GDRs, NVDRs & Certificates) and their root assets. Note that specific risk forecasts and specific return correlations for different share classes, e.g. 'A' and 'B', ordinary and preference shares, are still estimated empirically under the structural approach as these lines confer different rights to the owner and are in no way fungible.

The model user is free to choose the approach that best applies to their investment process and horizon. The structural approach is typically used by active portfolio managers who view related listings as being entirely fungible over longer horizons, i.e. many months, whereas the empirical approach is typically more suited to index tracking portfolios, where even small differences in how related listings trade on a day-on-day basis can significantly impact their performance.

Comparison of BFRE 1.0 and 2.0 in model estimation stages

Table 5: Comparison of BFRE 1.0 and 2.0 in model estimation stages

Model estimation stage	BFRE 1.0 vs. BFRE 2.0
Model universe construction	same
Factor exposures	1.0 models update exposures weekly[18], 2.0 models update exposures daily
Factor Returns	same for 1.0 and 2.0 Short Horizon model; Long horizon model does not include Reversal
Specific returns	same for 1.0 and 2.0 Short Horizon model; Long horizon model is different
Factor covariance matrices	different due to 2.0 enhancement
Specific risk	different due to 2.0 enhancement
Specific return correlation	same for 1.0 and 2.0 Short Horizon model; Long horizon model is different

[18] more recent models including BFRE 1.0 India, China, North American Healthcare, South Africa, Brazil have daily updates

Figure 20: contribution to analytical risk report in Aladdin, NAMR 1.0 model, 27 March 2020

Risk Group	Factor Level	Factor Vol	Credit Beta	Exposure	Stand-alone Risk	Risk Contribution
Equity		matrix...			2,167	2,167
NAMR Style		matrix...		1.82	2,200	2,182
NAMR Sector		matrix...		99.98	208	-49
NAMR Country		matrix...		99.98	89	23
Equity Specific		matrix...		0.02	154	11
Foreign Exchange		matrix...			1	0
FX Spot		matrix...			1	0
Total		**matrix...**			**2,167**	**2,167**

Figure 21: contribution to analytical risk report in Aladdin, NAMR 2.0 Short Horizon model, 27 March 2020

Risk Group	Factor Level	Factor Vol	Credit Beta	Exposure	Stand-alone Risk	Risk Contribution
Equity		matrix...			3,934	3,934
NA2S Style		matrix...		1.85	4,007	3,983
NA2S Sector		matrix...		99.98	371	-91
NA2S Country		matrix...		99.98	164	35
Equity Specific		matrix...		0.02	160	6
Foreign Exchange		matrix...			2	1
FX Spot		matrix...			2	1
Total		**matrix...**			**3,936**	**3,936**

3.3 Model Assumptions & Limitations

Model users should note the following assumptions and limitations of the BFRE risk models:

- All the standard assumptions of BRS' Covariance Matrix Estimation methodology apply
- Due to past vendor limitations the Chinese Stock Connect Securities were covered for risk estimation purposes but not for optimization purposes in BFRE 1.0. They are now covered for both purposes in the BFRE 2.0 models.
- The forecast horizon of the Short Horizon models is 1-month. This is reflected in different aspects of the model construction:
  - Style and industry factors are selected by assessing explanatory power using monthly cross-sectional regressions during the research period 1996-2013
  - Risk estimates are computed at a 1-month horizon taking into account daily serial correlations in Factor Returns and asset specific returns
  - BFRE 2.0 gamma scaling is based on monthly realised volatilities
  - Model bias statistics are evaluated using monthly standardised returns
- The forecast horizon of the Long Horizon models is 6-month:
  - Reversal factor is removed from Long Horizon models. Other style and industry factors inherit from Short Horizon models and do not have a separate factor selection process
  - Risk estimates are computed at a 6-month horizon taking into account daily serial correlations in Factor Returns and asset specific returns
  - BFRE 2.0 gamma scaling is based on quarterly realised volatilities
  - Model bias statistics are evaluated using 6-month standardised returns
- Industry, country and currency factor exposures are assigned using dummy variables, e.g. all assets with exposure to the same industry factor receive the same risk from that exposure. Heterogeneity in the industry block is captured via the granularity of the industry factors
- The currency factor exposures are always assigned based upon the country factor exposures
- Currency Factor Returns are truncated to remove outliers prior to the estimation of the factor covariance matrix. Daily currency returns are bounded by +/- 25%, and weekly currency returns are bounded by +/- 50%[19], for 2.0 models currency returns are bounded by +/- 50% for both daily and weekly models
- Factor Returns have zero correlation with asset specific returns, and specific returns from different issuers are unrelated and have zero correlation
- BFRE 2.0 factor risk gamma adjustment assumes co-movement in factor volatilities are captured by a common factor such that the cross-section of factor volatilities can be used to adapt individual factor volatility forecasts
- Specific return correlations are computed relative to a representative, root asset. Despite being a popular approach in industry, this model does not adequately capture all relationships between different listings of companies, which have more than one share class with derived securities linked to those share classes, e.g. Chinese MMA securities
- Dual-listed companies share the same fundamental data for both sister companies, and therefore have identical fundamental, style factor exposures. This treatment also extends to Royal Dutch Shell which retains aspects of the dual-listed structure after its unification

[19] Before March 2022 daily currency returns were bounded by +/- 8%, and weekly currency returns were bounded by +/- 20%

- The model should be used with care on portfolios covering assets not well represented in the estimation universe
- The model is updated daily and is delivered to all Aladdin clients on a 'T-2' basis

  Model users are referred to [13] for more details on the assumptions and limitations of fundamental multi-factor models.

4 Model Data & Inputs

This section describes the input data used by the BFRE risk models.

Item Description	Data Source(s)	Preparation & Quality Control	Limitations
Company Fundamentals	Worldscope*	Rolling 12-month ('RTM') logic applied to choose between annual, semi-annual and quarterly items Large differences reviewed in Data QC	Fundamental price ratios are normalized by Worldscope prices
Analyst Estimates	I/B/E/S*	Large differences reviewed in Data QC	
Market Data (e.g. returns, volumes)	Datastream	Roll-forward logic applied to stale/illiquid assets Returns reviewed as part of Data QC****	
Company Market Capitalisation	Datastream		Limited coverage; in-house logic applied as an alternative
Industry Classifications	Current: The Refinitiv Business Classification Historical**: MISCI S&P Global Industry Classification Schema ('GICS') and FTSE Russell Industry Classification Benchmark ('ICB')***	Cloning logic applied to ensure consistency across all listings within the same issuer	Current: overrides applied to expand coverage Historical**: overrides applied to separate composite assets into multi-sector sub-industry, and expand coverage
Asset Coverage / Universe	Bloomberg Equity BulkFeed & Datastream Equity	Coverage universe restricted to assets with sufficient market data	Securities on the same exchange with different traded currencies, e.g. Chinese MMA securities
Country Classifications	Preferred: Worldscope Fall-back: Datastream	Proxies applied to small jurisdictions, e.g. Panama -> USA	Overrides applied on an ad-hoc basis
FX / Risk Free Rate	Reuters	Large differences reviewed in Data QC	

*via Thomson Quantitative Analytics ('TQA')
**before February 2018
***BFRE UK model only
****IDC and Bloomberg market data are used as secondary source for Data QC

5 Model Testing

All BFRE 1.0 models were tested extensively using a history of data spanning several different market environments from 1996 to 2013. The section below provides a brief overview of the various diagnostics and backtests performed on the BFRE 1.0 models prior to their release in 2013, as well as the testing for BFRE 2.0 enhancements prior to the release in 2023.

5.1 BFRE 1.0 Model Estimation Diagnostics

The model estimation is a core component of the model construction process. Several standard diagnostics were appraised during this process to check the overall explanatory power of the models, individual factor efficacy and also multicollinearity:

- The R^2 quantifies the proportion of cross-sectional variation in asset returns explained by the set of common factors in the model. This is a standard measure of any factor model's overall explanatory power
- Individual factor efficacy is assessed using a history of t-statistics for each common factor by calculating the proportion of significant t-statistics over different periods. This metric serves as a good proxy for the persistence of individual factor effects. The majority of factors are significant more than 10% of time over the research history, with most well in excess of this threshold
- The final set of diagnostics are variance inflation factors. These help diagnose issues relating to multicollinearity between the style factors. If style exposures are too closely correlated then the regression procedure will encounter problems in apportioning the factor return between them. This can result in significant instability in the factor return estimates through time[20]. In order to guard against these effects, variance inflation factors were reviewed over the research history and were found to be well within suitable thresholds

5.2 BFRE 1.0 Assessment of Volatility Forecasts

A well-calibrated model should provide risk forecasts that are stable and accurate. One way of testing the accuracy of the risk forecasts is to compare the volatility of realised returns to forecast risks. This forms the basis of the bias statistic used to evaluate the BFRE models.

An exhaustive set of bias statistics was generated and reviewed over the entire research history. The tests themselves were conducted using an exhaustive set of portfolios:

- Pure factor portfolios for all common factors: markets, styles, industries, countries and currencies
- Market capitalisation-weighted estimation universes, and their industry and country based carve-outs
- Market capitalisation-weighted terciles of the estimation universe into large-cap, mid-cap and small-cap segments
- Individual stocks in the estimation universe
- Minimum variance portfolios
- Active portfolios

Furthermore, the final suite of test statistics included a significant number of genuine portfolios covering different geographical areas and market segments.

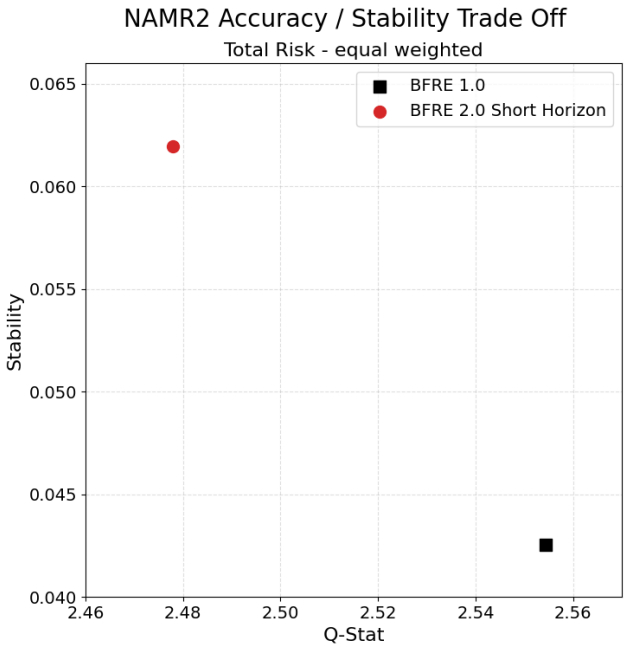
[20] In the most extreme case, where factor exposures are perfectly correlated, identification issues will exist causing the estimation process to fail

5.3 BFRE 2.0 Assessment of Volatility Forecasts

Forecast performance for BFRE 2.0 models is benchmarked against the corresponding BFRE 1.0 model. Forecast accuracy is assessed by Patton's Q-statistic[24][21] that accounts for both the bias and precision of volatility forecasts using realised returns[22] standardised by volatility forecast. Q-statistic penalises underprediction more than overprediction and attains a minimum value when the volatility forecast used to standardise returns matches the true latent volatility. Forecast stability is measured by the absolute percentage change in the risk forecast from month to month.

Figure 22 to 25 present the Q-stats and stability results comparing BFRE 2.0 NAMR Short Horizon model and Long Horizon model with BFRE 1.0 NAMR model over the full sample period, equal-weighted (Figure 22 and 23) and cap-weighted (Figure 24 and 25) asset-level over the NAMR estimation universe. The 2.0 Short Horizon model has significant gain in forecast accuracy while the more adaptive forecasts are inherently less stable than 1.0 model. The 2.0 Long Horizon model has improvements in both forecast accuracy and stability.

Figure 22: Q-stats and stability: BFRE 1.0 NAMR vs. BFRE 2.0 Short Horizon NAMR, 2005 - 2024



[21] Q-statistic is defined as $q_t = {z_t}^2 - \log {z_t}^2$ where z_t is the realised return standardised by volatility forecast at time t

[22] BFRE 2.0 Short Horizon models use monthly realised returns while Long Horizon models use semi-annual realised returns. This is consistent with the desired forecast horizon

Figure 23: Q-stats and stability: BFRE 1.0 NAMR vs. BFRE 2.0 Long Horizon NAMR, 2005 - 2024

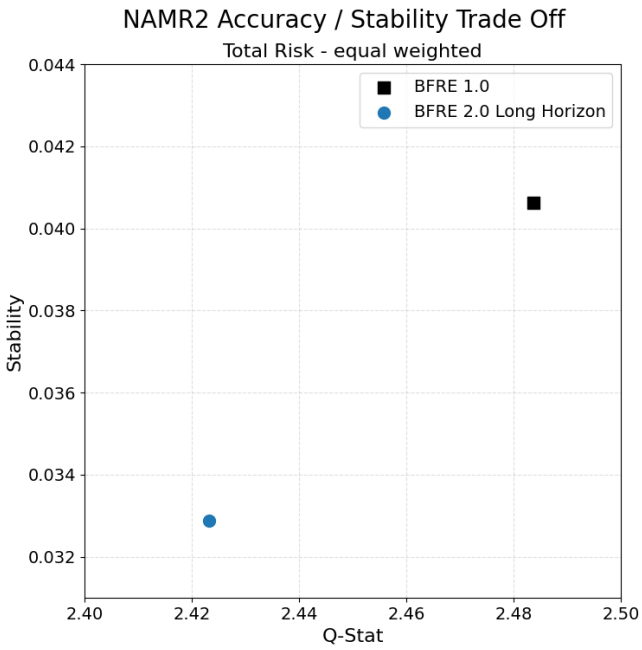


Figure 24: Q-stats and stability: BFRE 1.0 NAMR vs. BFRE 2.0 Short Horizon NAMR, 2005 - 2024

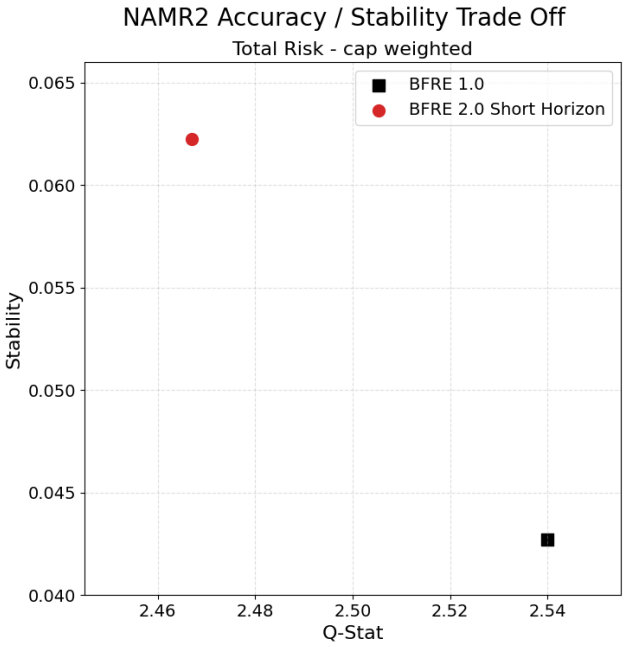
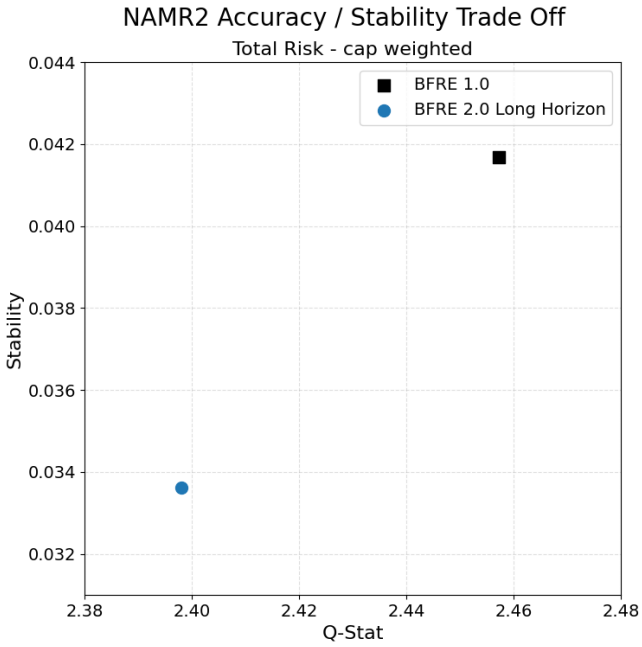


Figure 25: Q-stats and stability: BFRE 1.0 NAMR vs. BFRE 2.0 Long Horizon NAMR, 2005 - 2024



6 Model Outputs and Use

6.1 BFRE Models in Aladdin

There are several tools which provide access to BFRE-based analytics in Aladdin: the Green Package, PRT, Portfolio Construction (PfC) and Explore. To give an example, Figure 26 shows an Equity Daily Risk (EDR) report for a European Equity portfolio using the EMEA model. This report was generated using PRT, and provides an extremely intuitive, graphical description of the active bets being taken in this portfolio.

Top-line risk numbers are shown along the top of the report: Active Risk, Portfolio Beta, Portfolio Risk and Benchmark Risk. Active Risk is then decomposed along the different factor blocks in the model, shown in the pie chart: styles (including the Market factor), industries, countries and currencies. The report shows that the Active Risk is split equally between common factors and stock specific sources. Style and industry factors account for most of the common factor risks.

For each factor block, there is a corresponding bar chart (using a consistent colour) that shows the top contributing factors in that block – this displays the active exposure together with the contribution to active risk. For instance, within the style block (bottom-left), this portfolio has significant tilts towards high volatility, momentum driven, low-yielding, smaller stocks.

Information of this nature is extremely useful and can be fed back into the portfolio construction process and used to promote more informed discussions between portfolio managers and risk managers around the level and types of risks being taken in portfolios. The factor structure used in BFRE is particularly relevant here as using an intuitive, relevant set of factors that align with how portfolio managers structure their investment process promotes clearer understanding of portfolio risks, leading to more informed decision-making and better portfolio construction.

6.2 Model Production Process & Quality Controls

The flowchart 27 illustrates the daily production process for the model estimation. We retrieve data from various vendor sources[23] and apply QC to ensure that all model inputs are cleaned and suitable for model usage. The data content covered in the Data QC procedure includes

- Raw market data: prices, shares and total returns
- Derived market data: security-level market capitalisations, company-level market capitalisations and volumes

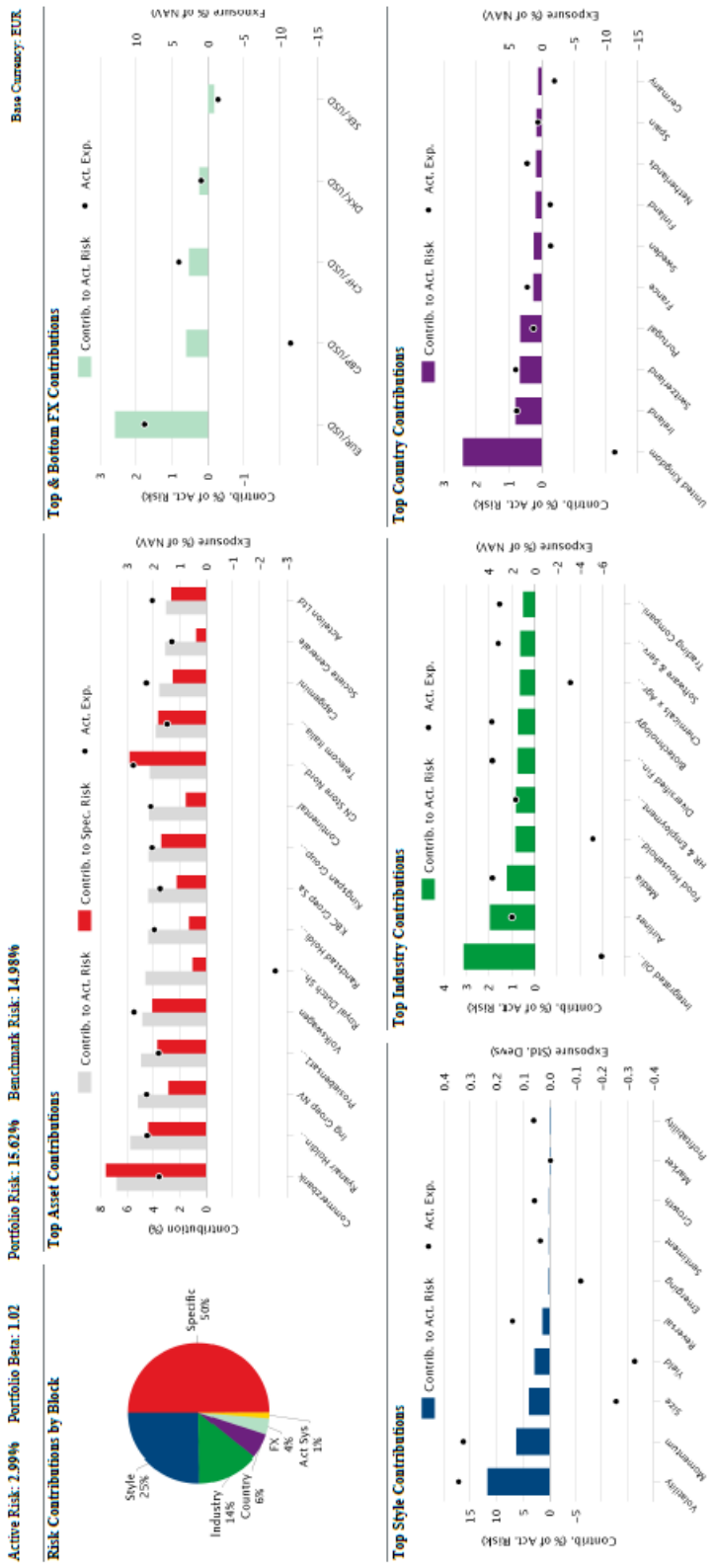
The procedure is exception-based and identifies missing returns and market capitalisations to ensure that all actively traded securities with non-zero trade volumes and prices have corresponding returns and market capitalisations. Additionally, the QC procedure validates securities with extreme returns and market capitalisations to ensure that any extreme values are accurate, and correctly reflect market events.

After the Data QC procedure, we proceed to the substyle standardization process which is then followed by style factor aggregation, as described in the Style Factors section. Once the factor exposures have been computed, the process of factor return estimation commences. The estimation procedure uses thinness corrections for country and industry factors where limited, or no data exists to reliably estimate those Factor Returns.

We impose the thin country/industry correction by adding a Bayesian prior, which in essence diverts the estimated country/industry return away from the sample factor return and towards a theoretical prior. In its purest sense this can be understood as capturing a true country/industry return rather than asset-specific returns. Estimated Factor Returns and specific returns are then used to compute a factor covariance matrix, and specific risk forecasts, respectively.

[23] See the Model Data & Inputs section for more details

Figure 26: a sample Equity Daily Risk (EDR) report in Aladdin

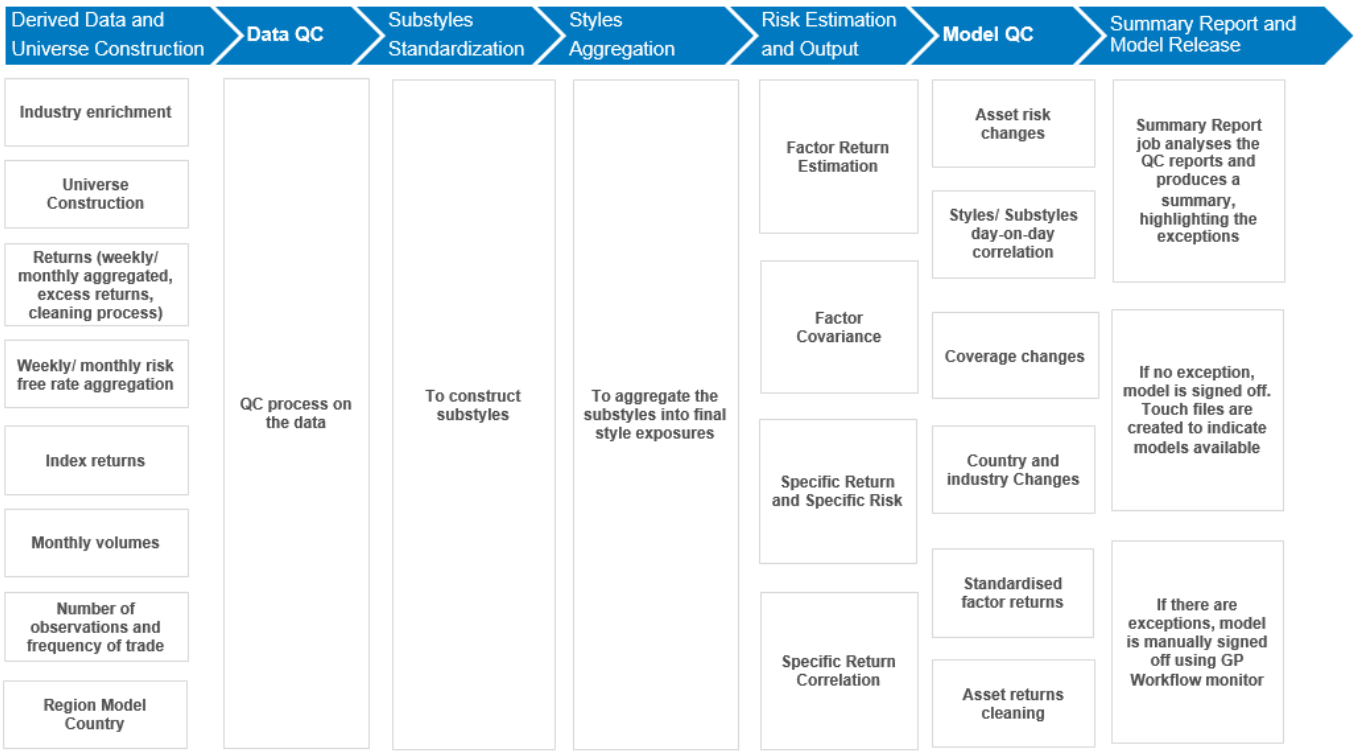


Prior to the models being released to Aladdin clients, the research team conducts a daily Model QC process that checks the quality of the model outputs. This is a separate process to the Data QC which is used to sanitise the raw inputs to the models. Instead, the Model QC process focuses on appraising

- the turnover in membership of the standardisation and estimation universes
- changes in factor exposures
- model estimation diagnostics and
- risk outputs: common factor volatilities, specific risk forecasts and specific return correlations

Daily Model QC comprises all checks that are performed on a daily basis to ensure that the models correctly reflect updates to Factor Returns, factor covariance matrices and incorporate new listings. The model universes, factor exposures, specific risk and specific return correlations are updated on a weekly basis[24] on Thursday to incorporate the data as of previous Wednesday market close, which necessitates additional checks on these items beyond those covered in the Daily Model QC.

Figure 27: BFRE Model Production Process



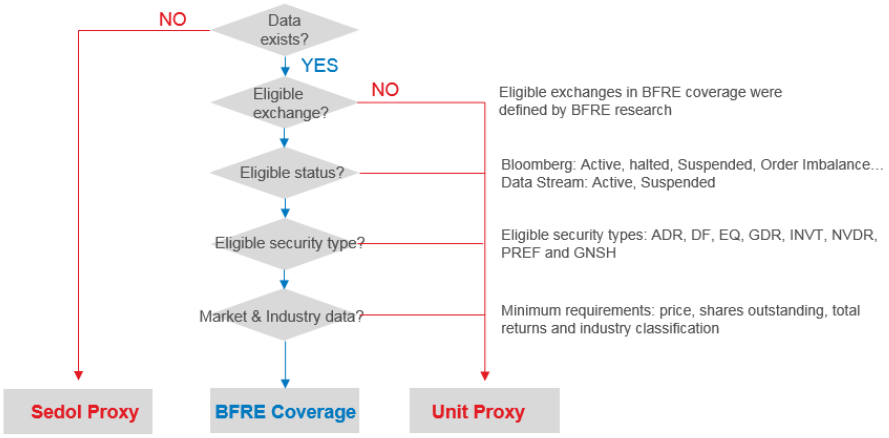
6.3 Proxying in Aladdin

As described in Figure 28, securities need to pass several criteria to be covered in the BFRE models. For securities not covered by the model, there are two types of proxy available depending on the availability of security meta-data. Typically, for a security with country and/or industry information, we apply a unit proxy that calculates a security's factor exposures using the average value from those in the same country and/or industry. For

[24] more recent models including BFRE 1.0 India, China, North American Healthcare, South Africa, Brazil, and BFRE 2.0 models have daily updates

a security with no such data, sedol proxying may be applied to map the security to another one with the same issuer.

Figure 28: BFRE Coverage Criteria and Proxy



7 Model Surveillance

The BFRE risk models are reviewed on an on-going basis as follows:

- Accuracy of volatility forecasts: this is assessed through the use of bias statistics which compare the level of forecast volatility against realised volatility for different portfolios. Bias statistics are based on a rolling window standard deviation of 36 monthly standardised returns, and exceptions are flagged using a 95% confidence interval
- Accuracy of tail risk forecasts: this is assessed based on the proportion of violations of 99% 1-day VaR over the previous 252 days[25]. VaR is calculated using BRS' various VaR methodologies: Analytical and Historical Var
- Benchmarking: BFRE risk forecasts for the metrics above are assessed against Aladdin's other equity risk models, namely STORM.

Exceptions are reviewed and logged on a monthly basis in line with current BRS model risk surveillance practices. Signs of persistently poor model performance over many periods would lead to further investigation by the model research team, and may lead to future model development.

The results of the model back-testing reviews are also presented to Aladdin clients in the Quarterly Aladdin Risk Model back-testing calls.

[25]This test is consistent with current UCITS guidelines for model evaluation, and follows the approach in Kupiec (1995)

8 Appendices

8.1 Style Factor Exposure Construction

Raw substyle values are built from input data according to a set of pre-specified Equations set out in Appendix 8.3. As part of this process, theoretical bounds may be placed on intermediate forms or set to missing (where a data error is suspected). Up to five transformations are applied to the raw substyles. Some of these consume model universe information, and hence these transformed substyles are model-specific. The transformations (in order of application) are:

1. **Cloning**
   There are two types of cloning according to whether a substyle is security or company specific. Securtiy cloning seeks to fill missing values by harvesting the value from a cross-listing, or from a root asset to a depositary receipt (and vice versa). Company cloning seeks to fill miss values by harvesting the value from the primary listing of the company. These procedures are typically used to handle new listings or securities of existing companies.

2. **Fill-Miss**
   This procedure seeks to cross-sectionally interpolate the substyle value by regressing it on a size factor (log market capitalisation), the market, country and industry factors[26]. For any asset that is missing a substyle value, its value is inferred using similar-sized assets in the same industry and the same country. This procedure is typically employed to handle listings of new companies.

3. **Huberisation**
   Huberisation is a refined version of the standard Winsorisation procedure for transforming substyle values to a common standardised scale, i.e. z-scores. The standardised values are referred to as exposures and take values between +/- 3 and are standardised to a square-root capitalisation mean of zero, with an equal-weighted standard deviation of one[27]. This procedure mitigates issues of clustering and destroyed rank information in the tails of the substyle (distribution) as compared to the standard Winsorisation procedure.

4. **Substyle Aggregation**
   Following the above transformations, substyles are aggregated to form style factors according to the definitions set out in Appendix 8.2. These values are then passed through the Huberisation procedure once more to ensure that style factor exposures are standardised to a common scale.

5. **Style Factor Exposure Cloning**
   Finally a cloning procedure is applied to ensure that fungible assets receive same style factor exposures. Assuming that no barriers exist, these assets should converge in traded price to eradicate any arbitrage opportunities. Thus, over longer horizons their returns are expected to converge, even if their shorter-term (observed) price journeys do not exactly match.

   This approach is applied to related listings such as Depositary Receipts (and their root assets) and cross-listed assets. In general style factor exposures are cloned from the more liquid (or primary) listing to the less liquid listing.

[26] For WRLD 1.0 there is an independent cross-sectional regresssion for each region
[27] The universe used for standardisation is similar to one used in the model estimation

8.2 Style Definitions

Style	Substyle Description	JAPN	APXJ	WRLD	USAM	EMKT
Volatility	Historical Beta (to Reginal Market Index)	0.20	0.34	0.25	0.34	0.34
	Hist. Beta x Hist. Sigma	0.20				0.33
	Cumulative Range - 12M	0.20	0.33	0.25	0.33	0.33
	Historical Sigma	0.20		0.25	0.33	
	Standard Deviation - 1Y	0.20	0.33	0.25		
Momentum	Relative Strength - 11M	0.50	0.50	0.50	0.50	0.50
	Historical Alpha	0.50	0.50	0.50	0.50	0.50
Size	Log Company Market Cap	0.34	0.34	0.34	0.50	0.25
	Log of Total Assets					0.25
	Broker Coverage*	0.33	0.33	0.33		0.25
	Log of Sales*	0.33	0.33	0.33	0.50	0.25
Reversal	Relative Strenght - 1M	1.00	1.00	1.00	1.00	1.00
Liquidity	Prop. of Active Trade days - 3M			0.17	0.20	
	Prop. of Active Trade days - 6M			0.17	0.20	
	Prop. of Active Trade days -12M			0.17	0.20	
	Price Pressure 1M*					0.34
	Price Pressure 3M*					0.33
	Return to Turnover*	1.00		0.16	0.20	
	Amihud Ratio - 1Y*		0.34	0.16		
	Growth of Trading Volume 12M					0.33
	Growth of Trading Volume 36M			0.17	0.20	
	Stock Turnover - 3M		0.33			
	Stock Turnover - 6M		0.33			
SmallCap	SmallCap Factor		1.00	1.00	1.00	1.00
Emerging/ Developed	Decoupling Beta to (iShares MSCI EM - SP500)	1.00		1.00		-1.00
Value	Book Value to Price	0.67	0.34	0.34	0.34	0.34
	Sales to Price	0.33	0.33	0.33	0.33	0.33
	Cash Flow to price		0.33	0.33	0.33	0.33
Profit	Return on Capital Employed				0.33	
	Retum on Assets			0.25	0.33	
	Interest Coverage			0.25	0.34	
	Cash Flowto Liabilities			0.25		
	Operating Profit Margin			0.25		
EamYield	Earnings to Price	0.34	0.34	0.34	0.34	0.34
	EBITDA to Enterprise Value	0.33	0.33	0.33	0.33	0.33
	Normalised Earnings to Price	0.33	0.33	0.33	0.33	0.33
DivYield	Dividend Yield			1.00	1.00	
Growth	Asset Growth Rate	0.34		0.25	0.34	
	Change in Assets	0.33		0.25	0.33	
	Growth of Total Sales	0.33		0.25	0.33	
	Variation in Capital Structure			0.25		
Sentiment	Beta on VIX (Regional)	1.00	1.00	1.00		
Leverage	Debt to Assets	0.34		0.50	0.50	
	Market Leverage	0.33				
	Balance Sheet Cash			-0.50		
	Book Leverage	0.33			0.50	
Oil	Crude Oil-Brent M+3	1.00				1.00
	UK Close U$/BBL (Regional)			1.00		
Quality	Equity Dilution		1.00			1.00

Note that the substyles with [*] are negated.

Style	Substyle Description	AUST	EMEA	UKIN	NAMR	CAND	LATC
Volatility	Historical Beta (to Reginal Market Index)	0.34	0.20	0.25	0.34	0.34	0.10
	Hist. Beta x Hist. Sigma		0.20	0.25			
	Cumulative Range — 12M	0.33	0.20	0.25	0.33	0.33	0.10
	Historical Sigma		0.20		0.33	0.33	0.40
	Standard Deviation — 1Y	0.33	0.20	0.25			0.40
Momentum	Relative Strength — 11M	0.50	1.00	0.50	0.50	0.50	0.50
	Historical Alpha	0.50		0.50	0.50	0.50	0.50
Size	Log Company Market Cap	0.34	0.25	0.25	0.75	0.34	1.00
	Log of Total Assets		0.25	0.25			
	Broker Coverage*	0.33	0.25	0.25		0.33	
	Log of Sales	0.33	0.25	0.25	0.25	0.33	
Reversal	Relative Strength -1M	1.00	1.00	1.00	1.00	1.00	1.00
Liquidity	Prop. of Active Trade days - 3M		0.20		0.10		
	Prop. of Active Trade days - 6M		0.20		0.10		
	Prop. of Active Trade days - 12M		0.20		0.10		
	Return to Turnover		0.20		0.10		
	Amihud Ratio - 1Y*		0.20		0.60	0.34	
	Stock Turnover - 3M			0.34		0.33	0.34
	Stock Turnover - 6M			0.33		0.33	0.33
	Stock Turnover - 12M			0.33			0.33
SmallCap	SmallCap Factor	1.00	1.00	1.00			
MidCap	MidCap Factor				1.00		
Emerging/ Developed	Decoupling Beta to (iShares MSCI EM - SP500)		1.00				
Value	Book Value to Price	0.34	0.34		0.34		
	Sales to Price	0.33	0.33	0.50	0.33		
	Cash Flow to Price	0.33	0.33	0.50	0.33		
Profit	Return on Capital Employed		0.25				
	Return on Equity		0.25		0.33		
	Return on Assets	0.33	0.25		0.33		
	Interest Coverage	0.34	0.25	0.50	0.34		
	Operating Profit Margin	0.33		0.50			
Earn Yield	Earings to Price		0.25		0.34	0.34	0.50
	EBITDA to Enterprise Value		0.25		0.33	0.33	
	Normalised Earnings to Price		0.25		0.33	0.33	0.50
Div Yield	Dividend Yield		0.25		1.00		
Growth	Asset Growth Rate		0.25		0.34	0.50	
	Change in Assets		0.25		0.33	0.50	
	Predicted Sales Growth		0.25		0.33		
	Growth of Total Sales		0.25				
	Variation in Capital Structure						
Sentiment	Beta on VIX (Regional)		1.00		1.00		
Leverage	Debt to Assets			0.34			1.00
	Market Leverage			0.33			
	Book Leverage			0.33			
Foreign Sensitivity	Foreign Sales					0.50	
	Foreign Assets					0.50	

Note that the substyles with [*] are negated.

Style	Substyle Description	CHNA	NAHC	SAFR	BRAZ	INDI
Volatility	Historical Beta (to Regional Market Index)	0.34	0.34	0.20	0.10	0.34
	Historical Beta x Historical Sigma			0.20		
	Cumulative Range - 12 Months	0.33	0.33	0.20	0.10	0.33
	Historical Sigma		0.33	0.20	0.40	
	Standard Deviation - 1 Year	0.33		0.20	0.40	0.33
Momentum	Relative Strength - 11 Months	0.50	0.50	1.00	0.50	0.50
	Historical Alpha	0.50	0.50		0.50	0.50
Size	Log Company Market Cap	0.34	0.75	0.25	1.00	0.34
	Log of Total Assets			0.25		
	Broker Coverage*	0.33		0.25		0.33
	Log of Sales	0.33	0.25	0.25		0.33
Reversal	Relative Strenght - 1 Month	1.00	1.00	1.00	1.00	1.00
Liquidity	Proportion of Active Trade days - 3 months		0.10			
	Proportion of Active Trade days - 6 months		0.10			
	Proportion of Active Trade days - 12 months		0.10			
	Return to Turnover*		0.10			
	Amihud Ratio - 1 Year*	0.34	0.60			0.34
	Stock Turnover - 3 months	0.33			0.34	0.33
	Stock Turnover - 6 months	0.33			0.33	0.33
	Stock Turnover - 12 months				0.33	
SmallCap	SmallCap Substyle	1.00				1.00
MidCap	MidCap Substyle		1 00			
Value	Book Value to Price	0.34	0.33	0.33		0.34
	Sales to Price	0.33	0.33	0.33		0.33
	Cash Flow to Price	0.33	0.34	0.34		0.33
Profitability	Return On Equity		0.33			
	Return on Assets		0.33			
	Interest Coverage		0.34			
Earnings Yield	Eamings to Price	0.34	0.34		0.50	0.34
	EBITDA to Enterprise Value	0.33	0.33			0.33
	Normalised Earnings to Price	0.33	0.33		0.50	0.33
Dividend Yield	Dividend Yield		1.00			
Growth	Asset Growth Rate		0.34	0.25	0.34	0.50
	Change in Assets		0.33	0.25	0.33	
	Predicted Sales Growth		0.33	0.25		
	Growth of Total Sales			0.25	0.33	0.50
Sentiment	Beta on VIX (Regional)	1.00	1.00			1.00
Leverage	Debt to Assets				1.00	
Quality	Equity Dilution	1.00				1.00

Note that the substyles with [*] are negated.

8.3 Substyle Definitions

Volatility

Historical Beta

The estimated slope coefficient $\hat{\beta}_i$ from an exponentially weighted univariate regression of asset returns r on a market index r^M:

$$r_{i.s} = \alpha_i + \beta_i r^M_{i,s} + \epsilon_{i,s} \tag{14}$$

All returns are weekly and in excess of the local risk-free rate.

The market index is defined as the market capitalisation-weighted Estimation Universe. BFRE WRLD 1.0 we use regional market indices as defined by the regional models and Japan, each asset is assigned the market index of the region of it's country exposure.

The exponential weighting function has a half-life of 52 weeks, and uses 5 years of weekly observations.

Historical Sigma

An equally-weighted standard deviation of the residuals in regression (14).

Beta $\times$ Sigma

The product of Historical Beta and Historical Sigma.

Cumulative Range – 12 months

The difference between the maximum and minimum cumulative log returns over the last 12 months:

$$\max_{s=0,\ldots,11}\left\{\sum_{u=t-s}^{t}\ln(1+r_{i,u})\right\} - \min_{s=0,\ldots,11}\left\{\sum_{u=t-s}^{t}\ln(1+r_{i,u})\right\} \tag{15}$$

All returns are total returns.

Standard Deviation – 1 year

An exponentially-weighted estimate of the standard deviation of daily total returns.

The exponential weighting function has a half-life of 180 days and uses 360 observations.

Size

Log of Company Market Capitalisation

The natural logarithm of company market capitalisation in USD.

Log of Total Assets

The natural logarithm of the average Total Assets A over the last 5 years:

$$\ln\left(\frac{1}{5}\sum_{s=t-4}^{t} A_{i,s}\right) \tag{16}$$

BFRE 1.0 models use total assets data in local currency. This treatment introduces inconsistent measures of size across companies in different geographies. From 2024 March BFRE 1.0 models use USD-denominated total assets data. BFRE 2.0 models use USD-denominated total assets data.

Broker Coverage

Broker Coverage is based on the number B of EPS FY1 estimates in the IBES database:

$$-\ln(1 + B_{i,t}) \tag{17}$$

Log of Sales

The natural logarithm of the average Total Sales S over the last 5 years:

$$\ln\left(\frac{1}{5}\sum_{s=t-4}^{t} S_{i,s}\right) \tag{18}$$

BFRE 1.0 models use total sales data in local currency. This treatment introduces inconsistent measures of sales across companies in different geographies. From 2024 March BFRE 1.0 models use USD-denominated total sales data. BFRE 2.0 models use USD-denominated total sales data.

Momentum

Relative Strength – 11 Months

The cumulative log excess return over the last 11 months[28] with a one month lag:

$$\ln\left(1+\prod_{s=t-11}^{t-1}(1+r_{i,s})-\prod_{s=t-11}^{t-1}(1+r_s^f)\right) \tag{19}$$

Historical Alpha

The Historical alpha from the exponentially weighted univariate regression (14).

Reversal

The log excess return over the previous month[29]:

$$\ln(1+r_{i,t}-r_t^f) \tag{20}$$

Liquidity

Proportion of Active Trade Days – 3 months, 6 months & 12 months

The proportion of active trade days over some specified window T. Active trade days are defined as those where the close price is not equal to the close price of the previous day, or where the traded volume is non-zero:

$$\frac{\sum_{s=t-T+1}^{t}\mathbf{1}_{(P_{i,s}\neq P_{i,s-1})\vee(V_{i,s}>0)}}{\sum_{s=t-T+1}^{t}\mathbf{1}_{(s\text{ is a trade day})}} \tag{21}$$

where $\mathbf{1}_X$ is a 0-1 indicator on the Boolean expression X and $T=$ number of days in the previous 3, 6 or 12 calendar months.

Return-to-Turnover

The average daily ratio of absolute total return to share turnover:

$$\frac{1}{T}\sum_{s=t-T+1}^{t}\frac{|r_{i,s}|}{V_{i,s}/N_{i,s}} \tag{22}$$

where $T=$ number of days in the last 360 with non-missing volume.

Amihud Ratio

The average daily ratio of absolute total return to value traded:

$$\frac{1}{T}\sum_{s=t-T+1}^{t}\frac{|r_{i,s}|}{V_{i,s}P_{i,s}} \tag{23}$$

where $T=$ number of days in the last 360 with non-missing volume.

An FX conversion is applied from local currency to USD to ensure comparability across all values.

[28] One month is measured as 22 working days
[29] One month is measured as 22 working days

Growth of Trading Volume – 12 months & 36 months

The ratio of the estimated slope coefficient from a regression of monthly traded volume on time, to the average Total Assets over the last T months:

$$\frac{\hat{\beta}_i}{\frac{1}{T}\sum_{s=t-T+1}^{t} V_{i,s}} \tag{24}$$

where $\hat{\beta}_i$ is estimated in the following regression:

$$V_{i,s} = \alpha_i + \beta_i s + \epsilon_{i,s} \tag{25}$$

where $s = t - T, \ldots, t$

Share Turnover – 3 months, 6 months & 12 months

The natural logarithm of the average daily ratio of traded volume to number of shares outstanding over some specified window T:

$$\ln\left(\frac{1}{T}\sum_{s=t-T+1}^{t}\frac{V_{i,s}}{N_{i,s}}\right) \tag{26}$$

where $T =$ number of days in the previous 3, 6 or 12 calendar months.

Price Pressure – 1 month & 3 months

The proportion of aggregated price pressure over aggregated total return for some specified window T. No trade days are defined as those where the close price is equal to the close price of the previous day, and the traded volume is zero.

The price pressure for asset i at time t over the previous T periods is defined as

$$\frac{\sum_{s=t-T+1}^{t}\delta_{i,s}|r_{i,s,\tau}|}{\sum_{s=t-T+1}^{t}|r_{i,s,\tau}|} \tag{27}$$

where $r_{i,s,\tau}$ is the estimate of asset return that would have occurred if the asset had traded, $r_{i,s}$ is the asset return, $\delta_{i,s}$ is an indicator the asset would have felt "price pressure" if traded, and τ is number of consecutive days the asset has not been trading as of time s

$$\delta_{i,s} = \begin{cases} 1, & \text{if } r_{i,s-1} = 0 \text{ or } v_{i,s-1} = 0 \\ 0, & \text{otherwise} \end{cases} \tag{28}$$

$$r_{i,s,\tau} = \begin{cases} r_{i,s}, & \text{if } \delta_{i,s} = 0 \text{ and } \delta_{i,s+1} = 0 \\ \prod_{k=0}^{\tau-1}(1 + r_{\text{market},s-k}) - 1, & \text{if } \delta_{i,s} \neq 0 \text{ or } \delta_{i,s+1} \neq 0 \\ 0, & \text{if model country is on holiday} \end{cases} \tag{29}$$

Earnings Yield

Earnings-to-Price

The ratio of Earnings-per-share EPS to month end price:

$$\frac{EPS_{i,s}}{P_{i,t}} \tag{30}$$

where $s \leq t$

EBITDA-to-Enterprise Value

The ratio of EBITDA to Enterprise Value EV:

$$\frac{EBITDA_{i,s}}{EV_{i,t}} \tag{31}$$

Normalised Earnings-to-Price

An Earnings-to-Price ratio where the numerator is replaced by the fitted value of the most recent EPS from a univariate regression of EPS on time:

$$\frac{\hat{\alpha}_i + \hat{\beta}_i s}{P_{i,t}} \tag{32}$$

where $\hat{\alpha}_i$ and $\hat{\beta}_i$ are estimated in the following regression:

$$EPS_{i,u} = \alpha_i + \beta_i u + \epsilon_{i,u} \tag{33}$$

The regression uses the last 5 years of EPS data.

Dividend Yield

The ratio of Dividends-per-share DPS to month end price:

$$\frac{DPS_{i,s}}{P_{i,t}} \tag{34}$$

where $s \leq t$

Value

Book Value-to-Price

The ratio of Book Value of Common Equity (per share) to month end price:

$$\frac{CE_{i,s}/N_{i,s}}{P_{i,t}} \tag{35}$$

where $s \leq t$

Sales-to-Price

The ratio of Net Sales/Revenues (per share) to month end price:

$$\frac{S_{i,s}/N_{i,s}}{P_{i,t}} \tag{36}$$

where $s \leq t$

Cash Flow-to-Price

The ratio of Funds From Operations (per share) to month end price:

$$\frac{FFO_{i,s}/N_{i,s}}{P_{i,t}} \tag{37}$$

where $s \leq t$

Growth

Asset Growth Rate

The ratio of the estimated slope coefficient from a regression of Total Assets A on time, to the average Total Assets over the last 5 years:

$$\frac{\hat{\beta}_i}{\frac{1}{5}\sum_{s=t-4}^{t} A_{i,s}} \tag{38}$$

where β_i is estimated in the following regression:

$$A_{i,s} = \alpha_i + \beta_i s + \epsilon_{i,s} \tag{39}$$

where $s = t-4, \ldots, t$

Change in Assets

The difference in the natural logarithm of Total Assets A_i over the previous two years:

$$\ln(A_{i,t}) - \ln(A_{i,t-1}) \tag{40}$$

Predicted Sales Growth

The percentage difference in the mean Sales forecast for FY1 and FY2:

$$\frac{FY2_{i,t} - FY1_{i,t}}{\frac{1}{2}(|FY2_{i,t}| + |FY1_{i,t})} \tag{41}$$

Growth of Total Sales

The ratio of the estimated slope coefficient from a regression of Total Sales S on time, to the average Total Sales over the last 5 years:

$$\frac{\hat{\beta}_i}{\frac{1}{5}\sum_{s=t-4}^{t} S_{i,s}} \tag{42}$$

where $\hat{\beta}_i$ is estimated in the following regression:

$$S_{i,s} = \alpha_i + \beta_i s + \epsilon_{i,s} \tag{43}$$

where $s = t-4, \ldots, t$

Variation in Capital Structure

The average 1-year change in various segments of the capital structure over the previous 4 years:

$$\frac{\frac{1}{4}\sum_{s=t-3}^{t}(|N_{i,s-1} - N_{i,s}|P_{i,s-1} + |LTD_{i,s-1} - LTD_{i,s}| + |PS_{i,s-1} - PS_{i,s}|)}{|CE_{i,t} + LTD_{i,t} + PS_{i,t}|} \tag{44}$$

where $s \leq t$

Profitability

Return on Capital Employed

The latest Return on Capital Employed provided by Worldscope.

Return on Equity

The ratio of Earnings-per-share EPS to average Book Value of Common Equity CE over the previous 2 years:

$$100 \times \frac{EPS_{i,t}}{\frac{1}{2}\left(\frac{CE_{i,t}}{N_{i,t}} + \frac{CE_{i,t-1}}{N_{i,t-1}}\right)} \tag{45}$$

Return on Assets

The latest Return on Assets provided by Worldscope.

Interest Coverage

The ratio of Operating Income OI to Interest Expense on Debt I:

$$\frac{OI_{i,t}}{I_{i,t}} \tag{46}$$

Cash Flow to Liabilities

The ratio of average Funds From Operations FFO over the previous 5 years, to Current Liabilities CL:

$$\frac{\frac{1}{5}\sum_{s=t-4}^{t} FFO_{i,s}}{CL_{i,t}} \tag{47}$$

Operating Profit Margin

The ratio of Operating Income OI to Net Sales S:

$$100 \times \frac{OI_{i,t}}{S_{i,t}} \tag{48}$$

Small-Cap

Small-Cap is a continuous function taking values in the interval [0,1], which identifies small cap stocks in the Estimation Universe:

$$SML_{i,t} = \begin{cases} e^{-\left(\frac{\ln(1+k_{i,t})-\mu_t}{\sigma_t}\right)^2}, & \text{if } k_{i,t} < \alpha_1 M_t \\ 1, & \text{if } k_{i,t} \geq \alpha_1 M_t \end{cases} \tag{49}$$

where

$$\begin{aligned} \mu_t &= \ln(1+\alpha_1 M_t) \\ \sigma_t &= \sqrt{\frac{-(\ln(1+\alpha_2 M_t)-\mu_t)^2}{\ln(\alpha_3)}} \\ M_t &= \text{number of assets in the Estimation Universe} \\ k_{i,t} &= \text{numerical rank assigned to the company market capitalisation relative to other companies in the Estimation Universe} \\ \alpha_1 &= 0.95, \alpha_2 = 0.75, \alpha_3 = 0.2 \end{aligned}$$

The functional form is designed to give exposure to smaller companies which reside in capitalisation deciles 8, 9 and 10.

Mid-Cap

The Mid-Cap factor for asset i, date t is defined as:

$$MID_{i,t} = e^{-\left(\frac{\ln(1+r_{i,t})-\mu_t}{\sigma_t}\right)^2} \tag{50}$$

where

$$\begin{aligned} \mu_t &= \ln(1+\alpha_1 N_t) \\ \sigma_t &= \sqrt{\frac{-(\ln(1+\alpha_2 N_t)-\mu_t)^2}{\ln(\alpha_3)}} \\ N_t &= \text{number of assets in the stadardisation universe at time } t \\ r_{i,t} &= \text{numerical rank of asset } i \text{ at time } t\text{, based on the company market capitalisation of asset } i \text{ relative to the standardisation universe at time } t\text{, defined below} \\ \alpha_1 &= 0.60,\ \alpha_2 = 0.54, \alpha_3 = 0.7 \end{aligned}$$

The Mid-Cap substyle assigns an exposure to companies in deciles 6 and 7.

Macro-economic Betas

The estimated slope coefficient $\hat{\gamma}_i$ from an exponentially weighted univariate regression of the residuals in regression (14) on the returns of the specified macro-economic factor f:

$$l\hat{\epsilon}_{i,s} = a_i + \gamma_i f_s + u_{i,s} \tag{51}$$

All returns are weekly and in excess of the local risk-free rate.
The exponential weighting function and data window match those used in regression 14.

Leverage

Debt-to-Assets

The latest Debt-to-Assets ratio provided by Worldscope.

Market Leverage

The ratio of Long-term Debt LTD and Preferred Stock PS to Market Capitalisation:

$$1 + \frac{LTD_{i,s} + PS_{i,s}}{N_{i,t} \times P_{i,t}} \tag{52}$$

where $s \leq t$

Balance Sheet Cash

The ratio of Balance Sheet Cash C to Total Assets A:

$$\frac{C_{i,t}}{A_{i,t}} \tag{53}$$

Book Leverage

The ratio of Long-term Debt LTD and Preferred Stock PS to the Book Value of Common Equity CE:

$$1 + \frac{LTD_{i,s} + PS_{i,s}}{CE_{i,t}} \tag{54}$$

Quality

Equity Dilution

The 1 year difference in log market capitalisation due to dilution of common equity:

$$-\ln\left[\frac{Capt_{i,t-1}(1 + r_{i,t})}{Capt_{i,t}}\right] \tag{55}$$

where r is the total return over the previous year.

Foreign Sensitivity

Foreign Sales

The Foreign Sales (FSAL) Sub-Style for company i, at financial period end t is defined as:

$$FSAL_{i,t} = \frac{FS_{i,t}}{S_{i,t}} \tag{56}$$

where $FS_{i,t}$ denotes the foreign sales for company i and $S_{i,t}$ the total assets of company i

Foreign Assets

The Foreign Assets (FASS) Sub-Style for company i, at financial period end t is defined as

$$FASS_{i,t} = \frac{FS_{i,t}}{A_{i,t}} \tag{57}$$

where $FS_{i,t}$ denotes the foreign assets for company i and $A_{i,t}$ the total assets of company i.

8.4 Inventory of All Substyles Investigated

This list shows an inventory of substyles that were investigated during the model construction process. This is a subset of the full list of 200+ which includes all variants, e.g. relative strength was constructed over different horizons: 1 month, 3 months, 6 months, 11 months (with a 1 month lag) and 12 months – which are counted as 5 different substyles.

Table 6: inventory of all substyles investigated

Style	Substyle
Macro Beta	Baltic Exchange Dry Index (BDI)
	Barclays Corporate BAA - AAA
	BARCLAYS EM WORLD INTL ISSUE
	Barclays High Yield B - CAA
	BARCLAYS US MORTGAGE BACKED SEC.
	CBOE SPX VOLATILITY VIX (NEW)
	CGBI WGBI EU ALL MATS ($)
	CGBI WGBI WORLD ALL MATS
	Crude Oil-Brent M+3 UK Close U$/BBL
	Crude Oil-WTI Spot Cushing U$/BBL
	iShares EM - S&P 500
	European Growth - Value
	S&P GSCI Agriculturl Total Return
	S&P GSCI Commodity Total Return
	S&P GSCI Copper Total Return
	S&P GSCI Energy Total Return
	S&P GSCI Gold Total Return
	S&P GSCI Industrial Metals Tot. Ret.
	S&P GSCI Livestock Total Return
	S&P GSCI Precious Metal Tot. Ret.
	S&P GSCI Zinc Total Return
Momentum	Log Price
	Relative Strength
Quality	Abnormal Accruals
	Accruals
	Asset Quality
	Asset Turnover
	Change in Accounts Payable
	Change in Accounts Receivable
	Change in Inventories
	Equity Dilution
	Net Operating Assets
Size	Age Using Availability of Total Assets
	Availability of Total Assets
	Availability of Earnings
	Broker Coverage
	Capitalisation
	Log Cap
	Market Cap Cubed
	Sales
	Total Assets
Financial Distress	Altmans Z-Score
	Loss Maker Dummy
	Loss Maker Duration
	Loss Maker Margin
	Loss Maker to Enterprise Value
	Loss Maker to Total Assets
	Ohlsons O-Score
Leverage	Book Leverage
	Debt to Assets
	Debt to Equity
	Balance Sheet Cash
	Current Ratio
	Market Leverage
Earnings Variability	Prop of Earnings that are Extraordinary Items
	Variability of Annual Earnings
	Variability of Cash Flow
	Variability of Earnings to Price
	Variation in Predicted Earnings
	Up vs Down EPS Revisions
	EPS Revisions
	Variation in Sales
Value	Book Value to Price
	Cash Flow to Price
	Sales to Price
	Sales to Enterprise Value
Profitability	Operating Profit Margin
	Return on Assets
	Return on Capital Employed
	Return on Equity
	Cash Flow to Liabilities
	Interest Coverage
Earnings Yield	Earnings to Price
	EBITDA to Enterprise Value
	Normalised Earnings to Price
	Negligible Earnings
Dividend Yield	Dividend Yield
	Extremely Low Yield
Growth	Asset Growth Rate
	Change in Assets
	Delta Earnings
	Normal Payout
	Earnings Growth Rate
	Growth of Total Sales
	Predicted Sales Growth
	Predicted Earnings Growth
	Variation in Capital Structure
Liquidity	Amihud Ratio
	Growth of Trading Volume
	Proportion of Active Trade Days
	Return to Turnover
	Stock Turnover
Cap Segments	Standardisation Universe Deciles
	Small Cap
	Mid Cap
Volatility	Cumulative Range
	Historical Sigma
	Standard Deviation
	Beta x Sigma
	Historical Beta
	Historical Alpha
Random	Random Substyle
Labour Intensity	Net Plant to Gross Plant
	Plant to Common Equity
Foreign Sensitivity	Currency Exposure
	Foreign Assets
	Foreign Operating Income
	Foreign Sales

8.5 Industry Schemas

The following Tables show the number of assets and market capitalisation (as of December 2024) for all BFRE industry factors.

Table 7: NAMR industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDIAUTOCMP	Automobiles & Components	23	1622
CDIDISTRET	Distrib & Speciality Retail	45	1196
CDIDIVSVCS	Diversified Consumer Services	6	39
CDIDURXHMB	Cons Dur & Apparel x Homebuild	37	425
CDIHMBLD	Homebuilding	15	215
CDIHOTLEIS	Hotels & Leisure	28	441
CDIMEDIA	Media	28	554
CDIRESTRNT	Restaurants	21	573
CDIRETMULT	Multiline Retail	7	123
CDIRETNET	Internet & Catalog Retail	7	2915
CSTALCTOB	Alcohol & Tobacco	9	377
CSTFOODSD	Food Hsehld Pers Prod S Drinks	53	1674
CSTFOODSTA	Food & Staples Retailing	23	1401
EGYEQ	Energy Equipment & Services	29	187
EGYEXP	Fuels Exploration & Prod	106	1731
EGYREFSTR	Oil & Gas Ref Mkt Trans & Stor	25	762
FINCAPMKT	Capital Markets	68	2470
FINCOMBANK	Commercial Banks	64	2343
FINCONSFIN	Consumer Finance	10	346
FINLIFREIN	Life Hlth Multiline Ins & Rein	26	479
FINMTG	Thrifts & Mortgage Finance	11	110
FINMULTSEC	Multi-Sector Holdings	1352	5537
FINPROPINS	Property Casualty Ins & Brkrs	34	1035
FINREAL	Real Estate	130	1488
FINSPECOTH	Spec & Other Div Fin Svcs	8	46
HLCBIOTECH	Biotechnology	64	782
HLCEQSUPLS	HC Eq Sup Life Sci Tools Svcs	57	1777
HLCMGDHLC	Managed Health Care	8	709
HLCPHARMA	Pharmaceuticals	28	2295
HLCPROVSVC	Health Care Providers & Svcs	39	341
INDAIRDEF	Aerospace & Defense	35	1154
INDAIRLINE	Airlines	9	123
INDBLDPRD	Building Products	21	367
INDCOMPRO	Commercial & Prof Services	38	585
INDCONENG	Construction & Engineering	23	220
INDCONMACT	Ind Mach Cong Trade Co & Dist	77	1505
INDELECEQ	Electrical Equipment	19	406
INDTRAXAIR	Transportation x Airlines	29	743
ITCCOMMEQ	Communications Equipment	12	525
ITCCOMPOFF	Comp Peripherals & Office Eq	13	4112
ITCELECEQ	Elec Equip Instr & Components	27	335
ITCITSERV	IT Services	69	3000
ITCNET	Internet Software & Services	23	6797
ITCSEMICEQ	Semiconductors & Equipment	47	6038
ITCSOFTW	Software	109	6416
MATALUDIV	Alum Div Metals & Mining	59	168
MATCHEM	Chemicals	41	768
MATCONSTM	Construction Materials	6	93
MATGOLDPRE	Gold Precious Metals Minerals	62	173
MATPAK	Containers & Packaging	16	135
MATPAPER	Paper & Forest Products	8	20
MATSTEEL	Steel	11	95
TELTELECOM	Telecommunication Services	18	885
UTLUTIL	Utilities	70	1464

Table 8: EMEA industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDIAUTOS	Automobiles	8	207
CDICOMPNT	Auto Components	20	78
CDIDIVSVCS	Diversified Consumer Services	3	2
CDIGAMING	Casinos & Gaming	13	43
CDIHHLEXHB	Hshld Dur x Hmbuild Leisure Eq	20	46
CDIHMBLD	Homebuilding	14	35
CDIHOTREST	Hotels Restaurants & Leisure	21	160
CDIMEDIA	Media	44	245
CDIRETAIL	Retailing	57	363
CDITXTLUX	Textiles Apparel Luxury Goods	26	899
CSTBEVTOB	Beverages & Tobacco	19	447
CSTFOODHH	Food Household & Personal Prod	58	962
CSTFOODSTA	Food & Staples Retailing	24	156
EGYEQ	Energy Equipment & Services	14	24
EGYEXPSTR	Fuels Exp Prod Storage & Trans	33	79
EGYINT	Integrated Oil & Gas	10	560
EGYREF	Oil & Gas Refining & Marketing	7	22
FINDIV	Diversified Financials	78	548
FINDIVMTG	Div Banks Thrifts & Mort Fin	94	1461
FINLIFREIN	Life Hlth Multiline Ins & Rein	39	746
FINMULTSEC	Multi-Sector Holdings	589	727
FINPROPINS	Property Casualty Ins & Brkrs	11	76
FINREAL	Real Estate	113	302
FINREG	Regional Banks	2	18
HLCBIOTECH	Biotechnology	28	184
HLCEQSVCLS	HC Equip Life Sci Tools & Svcs	58	455
HLCPHARMA	Pharmaceuticals	39	1691
INDAIRDEF	Aerospace & Defense	18	345
INDAIRFRGT	Air Freight & Logistics	7	107
INDAIRLINE	Airlines	11	64
INDBLDPRD	Building Products	19	140
INDCOMSUP	Commercial Services & Supplies	36	246
INDCONENG	Construction & Engineering	60	240
INDCONGLOM	Industrial Conglomerates	15	82
INDELECEQ	Electrical Equipment	20	484
INDHR	HR & Employment Services	7	18
INDMACHINE	Machinery	80	695
INDMARINE	Marine	16	120
INDROAD	Road & Rail	7	13
INDTRANINF	Transportation Infrastructure	14	83
INDTRDDIST	Trading Companies & Distrib	19	132
ITCHARDW	Technology Hardware & Equip	42	190
ITCSEMICEQ	Semiconductors & Equipment	24	449
ITCSOFTSVC	Software & Services	115	885
MATAGR	Fertilizers & Agriculture Chem	5	10
MATALUDIV	Alum Div Metals & Mining	19	256
MATCHEMXAG	Chemicals x Agrochemicals	43	376
MATCONSTM	Construction Materials	15	117
MATGOLD	Gold	6	30
MATPAK	Containers & Packaging	13	51
MATPAPER	Paper & Forest Products	13	59
MATPRECMET	Precious Metals & Minerals	5	18
MATSTEEL	Steel	14	29
TELTELECOM	Telecommunication Services	32	436
UTLUTIL	Utilities	66	670

Table 9: UKIN industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CSGFOODBEV	Food and Beverage	13	215
CSGHOMECNS	Home Construction	8	31
CSGPERSHHG	Personal & Household Goods	8	197
CSVFOODRUG	Food & Drug Retailers	2	39
CSVGENRTLR	General Retailers	21	66
CSVMEDIA	Media	10	46
CSVTRAVEL	Travel & Leisure	16	158
FINBANKS	Banks	7	339
FININVTRST	Investment Trusts	344	297
FINLIFE	Life Insurance	8	77
FINNONLIFE	Nonlife Insurance	6	35
FINREAL	Real Estate	18	55
FINSERVICE	Financial Services	25	142
HLCEQSVC	Health Care Equipment &Service	5	20
HLCPHARBIO	Pharmaceuticals & Biotechnology	6	322

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
INDAIRDEF	Aerospace & Defense	5	109
INDCONMAT	Construction & Materials	12	16
INDGENELEQ	General Ind &Elec &Engineering	19	71
INDSUPPORT	Support Services	25	282
INDTRANSPT	Industrial Transportation	3	7
MATCHEM	Chemicals	4	11
MATRESOURC	Basic Resources	11	234
OILEQUIP	Oil Equipment Services &Distri	2	1
OILGASALT	Oil &Gas producers &Alt Energy	8	276
TECHARDEQP	Technology Hardware &Equipment	4	4
TECSOFTCMP	Software & Computer Services	9	29
TELTELECOM	Telecommunications	5	50
UTLELEC	Electricity	2	25
UTLGASWTR	Gas & Water & Multiutilities	6	88

Table 10: JAPN industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDIAUTOCMP	Automobiles & Components	53	591
CDICONSVCS	Consumer Services	54	101
CDIDIST	Distributors	4	6
CDIHHDUR	Household Durables	19	211
CDILEISEQ	Leisure Equipment & Products	11	45
CDIMEDIA	Media	25	52
CDIRETMULT	Multiline Retail	12	44
CDIRETNET	Internet & Catalog Retail	5	22
CDIRETSPEC	Specialty Retail	37	180
CDITXTLUX	Textiles Apparel Luxury Goods	16	36
CSTALCTOB	Alcohol & Tobacco	5	86
CSTXALCTOB	Cons Staples x Alcohol & Tob	110	305
EGYENERGY	Energy	13	53
FINBANK	Banks	53	455
FINCAPMKT	Capital Markets	17	58
FINDIVSVCF	Div Fin Svcs Consumer Fin	21	70
FININS	Insurance	9	208
FINREAL	Real Estate	89	223
HLCEQSVC	Health Care Equipment & Svcs	38	140
HLCPHARBIO	Pharmaceuticals Biotechn	34	294

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
INDBLDPRD	Building Products	13	62
INDCOMPRO	Commercial & Prof Services	54	201
INDCONENG	Construction & Engineering	60	91
INDCONGLOM	Industrial Conglomerates	2	1
INDELECEQ	Electrical Equipment	23	128
INDMACHINE	Machinery	93	305
INDTRAN	Transportation	49	203
INDTRDDIST	Trading Companies & Distrib	32	303
ITCCOMMEQ	Communications Equipment	6	13
ITCCOMP	Computers & Peripherals	16	95
ITCELECEQO	Elec Eq Instr Comp & Off Elec	51	343
ITCSEMICEQ	Semiconductors & Equipment	32	224
ITCSOFTSVC	Software & Services	102	380
MATCHEM	Chemicals	77	261
MATCONSTM	Construction Materials	8	8
MATMINXST	Metals & Mining x Steel	12	18
MATPAK	Containers & Packaging	5	7
MATPAPER	Paper & Forest Products	5	8
MATSTEEL	Steel	21	49
TELTELECOM	Telecommunication Services	7	313
UTLUTIL	Utilities	20	69

Table 11: LATC industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDICONSDIS	Consumer Discretionary	29	42
CSTALCTOB	Alcohol & Tobacco	7	53
CSTXALCTOB	Cons Staples x Alcohol & Tob	33	231
EGYENERGY	Energy	12	204
FINBANK	Banks	38	329
FINDIV	Diversified Financials	17	70
FININS	Insurance	12	32
FINMULTSEC	Multi-Sector Holdings	32	18
FINREAL	Real Estate	21	35
HLCHLC	Health Care	10	18

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
INDINDUST	Industrials	44	157
ITCIT	Information Technology	5	5
MATCHEM	Chemicals	6	9
MATCONSTM	Construction Materials	6	14
MATMIN	Metals & Mining	29	149
MATPAK	Containers & Packaging	3	14
MATPAPER	Paper & Forest Products	3	17
TELTELECOM	Telecommunication Services	7	65
UTLUTIL	Utilities	54	173

Table 12: APXJ industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDIAUTOCMP	Automobiles & Components	88	177
CDICONSVCS	Consumer Services	104	135
CDIHHDUR	Household Durables	33	56
CDILEISEQ	Leisure Equipment & Products	24	19
CDIMEDIA	Media	84	50
CDIRETAIL	Retailing	99	202
CDITXTLUX	Textiles Apparel Luxury Goods	84	60
CSTALCTOB	Alcohol & Tobacco	9	37
CSTXALCTOB	Cons Staples x Alcohol & Tob	168	217
EGYENERGY	Energy	65	232
FINBANK	Banks	55	919
FINDIV	Diversified Financials	130	431
FININS	Insurance	36	334
FINMULTSEC	Multi-Sector Holdings	147	152
FINREALMGT	Real Estate Management & Dev	146	208
FINREIT	Real Estate Investment Trusts	98	175
HLCHLC	Health Care	395	455
INDBLDPRD	Building Products	18	9
INDCOMPRO	Commercial & Prof Services	67	56
INDCONGLOM	Industrial Conglomerates	17	86
INDCONMACH	Construction Farm Mach Trucks	36	71
INDDEFENG	Aerospace & Def Const & Eng	122	112
INDELECEQ	Electrical Equipment	96	170
INDINDMACH	Industrial Machinery	101	44
INDTRAN	Transportation	98	259
INDTRDDIST	Trading Companies & Distrib	31	37
ITCCOMMEQ	Communications Equipment	49	145
ITCCOMP	Computers & Peripherals	98	434
ITCELECEQO	Elec Eq Instr Comp & Off Elec	179	267
ITCITSERV	IT Services	92	47
ITCSEMICON	Semiconductors	227	1249
ITCSEMIEQ	Semiconductor Equipment	104	77
ITCSOFTNET	Software & Internet Services	131	562
MATAGR	Fertilizers & Agriculture Chem	7	4
MATALU	Aluminum	5	1
MATCHEMIND	Chemicals & Industrial Gases	100	91
MATCONSTM	Construction Materials	34	32
MATDIVPREC	Div & Prec Metals & Mining	62	289
MATGOLD	Gold	30	42
MATPAKPAP	Containers Packaging & Paper	35	13
MATSPECHEM	Specialty Chemicals	45	22
MATSTEEL	Steel	62	92
TELTELECOM	Telecommunication Services	28	390
UTLUTIL	Utilities	54	149

Table 13: WRLD industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDIAUTOCMP	Automobiles & Components	195	2677
CDICONSVCS	Consumer Services	258	1507
CDIDISTNET	Dist Net Catalog Multi Retail	109	3360
CDIHHDUR	Household Durables	111	630
CDILEISEQ	Leisure Equipment & Products	56	163
CDIMEDXPUB	Media x Broadcast & Publishing	139	721
CDIPUBLISH	Broadcasting & Publishing	43	181
CDIRETSPEC	Specialty Retail	182	1723
CDITXTLUX	Textiles Apparel Luxury Goods	153	1269
CSTALCTOB	Alcohol & Tobacco	45	959
CSTXALCTOB	Cons Staples x Alcohol & Tob	485	4997
EGYINTSTR	Fuels Refining Storage & Trans	161	2700
EQYEQEXP	Energy Equip Oil & Gas Exp	166	1169
FINCONSFIN	Consumer Finance	48	423
FINDIVBANK	Diversified Banks	303	5561
FINDIVXCON	Div Fin x Con Fin x Multi Hlds	314	3640
FININSBROK	Ins Brkrs Reins Property & Cas	80	1516
FINLIFEINS	Life Health & Multiline Ins	93	1408
FINMULTSEC	Multi-Sector Holdings	2297	6440
FINREALMGT	Real Estate Management & Dev	327	698
FINREGMTG	Reg Banks Thrifts & Mort Fin	19	116
FINREIT	Real Estate Investment Trusts	310	1782
HLCBIOLIFE	Biotechnology & Life Sciences	250	1271
HLCEQSUP	Health Care Equip & Supplies	207	2326
HLCPHARMA	Pharmaceuticals	212	4367
HLCPROVTEC	Health Care Providers & Tech	134	1205
INDAIRDEF	Aerospace & Defense	77	1555
INDBLDPRD	Building Products	73	581
INDCOMPRO	Commercial & Prof Services	208	1112
INDCONENG	Construction & Engineering	257	637
INDCONGLOM	Industrial Conglomerates	45	419
INDELECEQ	Electrical Equipment	160	1224
INDMACHTRD	Machinery & Trading Companies	471	2876
INDTRAN	Transportation	265	1789
ITCCOMMEQ	Communications Equipment	74	767
ITCCOMP	Computers & Peripherals	138	4667
ITCDATAOUT	Data Processing & Outsourcing	69	1966
ITCELECEQO	Elec Eq Instr Comp & Off Elec	287	1027
ITCSEMICEQ	Semiconductors & Equipment	438	8048
ITCSOFTXDA	Software & Svcs x Data Proc	584	16089
MATALUPRE	Alum Div & Prec Metals & Min	184	827
MATCHEM	Chemicals	328	1549
MATCONSTM	Construction Materials	73	268
MATGOLD	Gold	88	236
MATPAK	Containers & Packaging	61	215
MATPAPER	Paper & Forest Products	40	107
MATSTEEL	Steel	126	365
TELTELECOM	Telecommunication Services	96	2096
UTLUTIL	Utilities	273	2534

Table 14: EMKT industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDIAUTOCMP	Automobiles & Components	175	802
CDICONSVCS	Consumer Services	126	142
CDIHHLEI	Household & Leisure	77	191
CDIHMBLD	Homebuilding	9	4
CDIMEDIA	Media	100	98
CDIRETAIL	Retailing	145	290
CDITXTLUX	Textiles Apparel Luxury Goods	143	239
CSTBEVERAG	Beverages	48	209
CSTFOODPRO	Food products	255	376
CSTFOODSTA	Food & Staples Retailing	66	299
CSTHHPRO	Household & personal products	64	150
CSTTOB	Tobacco	11	103
EGYEQ	Energy Equipment & Services	37	78
EGYEXP	Fuels Exploration & Prod	53	540
EGYINT	Integrated Oil & Gas	35	847
EGYREF	Oil & Gas Refining & Marketing	26	46
FINBKASEAN	Banks APXJ x China India Taiwan Korea	76	388
FINBKBRAZL	Banks Brazil	16	216
FINBKCHINA	Banks China	13	1385
FINBKEUXRU	Banks Europe x Russia Greece	15	94
FINBKINDIA	Banks India	43	589
FINBKKOREA	Banks Korea	11	78
FINBKLAXBR	Banks LATC x Brazil	22	113
FINBKMEAGR	Banks MEA x Turkey + Greece	37	321
FINBKTAIWA	Banks Taiwan	18	134
FINBKTURKY	Banks Turkey	12	102
FINDIV	Diversified Financials	228	858
FININS	Insurance	93	908
FINREIT	Real Estate Investment Trusts	88	68
FINREMAPXJ	Real Estate Mgmt & Dev APXJ x China India Taiwan	109	104
FINREMCHN	Real Estate Mgmt & Dev China	15	78
FINREMEMEA	Real Estate Mgmt & Dev EMEA	45	109
FINREMIND	Real Estate Mgmt & Dev India	34	102
FINREMLATC	Real Estate Mgmt & Dev LATC	14	21
FINREMTWN	Real Estate Mgmt & Dev Taiwan	48	31
HLCBIOTECH	Biotechnology	111	204
HLCEQSVC	Health Care Equipment & Svcs	160	208
HLCPHARBIO	Biotechn	193	393
INDAIRDEF	Aerospace & Defense	29	124
INDAIRFRGT	Air Freight & Logistics	29	43
INDAIRLINE	Airlines	31	126
INDBLDPRD	Building Products	47	40
INDCOMPRO	Commercial & Prof Services	67	50
INDCONENG	Construction & Engineering	193	322
INDCONGLOM	Industrial Conglomerates	50	470
INDELECEQ	Electrical Equipment	153	448
INDMACHINE	Machinery	223	320
INDMARINE	Marine	38	112
INDTRANINF	Transportation Infrastructure	92	256
INDTRDDIST	Trading Companies & Distrib	33	55
ITCHARDW	Technology Hardware & Equip	344	929
ITCITSERV	IT Services	142	547
ITCNET	Internet Software & Services	38	1437
ITCSEMICON	Semiconductors	245	1338
ITCSEMIEQ	Semiconductor Equipment	103	81
ITCSOFTW	Software	100	167
MATALUDIV	Alum Div Metals & Mining	70	279
MATCHEM	Chemicals	298	445
MATCONAPXJ	Construction Materials APXJ x Chin	46	44
MATCONCHN	Construction Materials China	4	22
MATCONEMEA	Construction Materials EMEA	19	18
MATCONIND	Construction Materials India	25	130
MATCONLATC	Construction Materials LATC	6	14
MATGOLDPRE	Gold Precious Metals Minerals	20	125
MATPAK	Containers & Packaging	40	31
MATPAPER	Paper & Forest Products	27	37
MATSTEEL	Steel	149	298
TELTELECOM	Telecommunication Services	75	958
UTLUTIL	Utilities	226	1009

Table 15: AUST industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDICONSDIS	Consumer Discretionary	51	157
CSTCONSTAP	Consumer Staples	14	58
EGYCOFUEL	Coal & Consumable Fuels	14	18
EGYOILGAS	Oil & Gas	11	53
FINBANKS	Banks	11	388
FINDIV	Diversified Financials	44	135
FININS	Insurance	8	47
FINREAL	Real Estate	39	112
HLCHLC	Health Care	32	155
INDITC	Industrials & IT	81	193
MATDIVMMM	Diversified Metals & Mining	53	310
MATGOLD	Gold	26	41
MATXDIVMET	Materials ex Metals & Mining	10	18
TELUTL	Telecom & Utilities	8	59

Table 16: CAND industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDICONSDIS	Consumer Discretionary	28	115
CSTCONSTAP	Consumer Staples	19	113
EGYDRILREXP	Oil & Gas Equipt Drilling Refining and Explo	58	150
EGYSTORCOF	Integ Oil & Gas Storage Coal and Cons Fuels	26	374
FINDIVFBANK	Banks & Diversified Financials	32	552
FININS	Insurance	10	219
FINREAL	Real Estate	43	73
HLCHLC	Health Care	26	18
INDINDUST	Industrials	47	314
ITIT	IT	29	137
MATALDIVMET	Alum Diversified Met & Steel	59	105
MATGOLD	Gold	49	111
MATPRECMM	Precious Metals & Mineral	10	7
MATXDIVMET	Materials ex Metals & Mining	19	53
TELTELECOM	Telecom	6	76
UTLUTIL	Utilities	15	81

Table 17: USAM industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDIAUTOCMP	Automobiles & Components	19	1607
CDICAS	Casino & Gaming	10	109
CDIDIVCS	Diversified Consumer Services	6	39
CDIDURXHOM	Consumer Durables ex Homebld	31	407
CDIHOMEBLD	Homebuilding	15	215
CDIHTLLEIS	Hotel Resort Cruise & Leisure Facilities	17	331
CDIINTRET	Internet & Catalog Retail	7	2915
CDIMEDIA	Media	23	543
CDIREST	Restaurants	15	536
CDISPECRET	Specialty Multiline Retails & Distributors	46	1287
CSTALCTOB	Alcohol & Tobacco	14	958
CSTFHHPERS	Food Household & Personal Products	39	1068
CSTFOODRET	Food & Staples Retailing	13	1315
EGYEQUIP	Energy Equipment & Services	16	178
EGYOGEXP	Oil & Gas Exploration & Production	34	566
EGYOGINT	Integrated Oil & Gas	23	1398
EGYOGREF	Oil & Gas Refining & Marketing	3	13
FINBANKS	Banks	62	1951
FINDIV	Diversified Financials	67	2811
FINLIFEINS	Life Health & Multi-line Insurance	16	285
FINMRGREIT	Mortgage REIT's	5	34
FINMRGREIT	Mortgage REITs	5	34
FINMULTSEC	Multi-Sector Holdings	480	5450
FINPROPINS	Property & Casualty Insurance	34	1010
FINREAL	Real Estate	82	1381

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
HLCBIOTECH	Biotechnology	62	782
HLCFAC	Health Care Facilities	20	278
HLCLIFESCI	Life Sciences Tools & Services	1	26
HLCMNG	Managed Health Care	8	709
HLCOTHER	Other Healtcare Equipment & Services	59	1801
HLCPHARMA	Pharmaceuticals	20	2290
INDAIR	Airlines	6	115
INDAIRDEF	Aerospace & Defense	28	1128
INDCOMPRO	Commercial & Professional Services	29	509
INDCONENG	Construction & Engineering	89	1809
INDELECTRD	Electric Equipment & Trading Co	31	632
INDTRNXAIR	Transportation ex Airlines	22	598
ITCCOMEQP	Communications Equipment	10	525
ITCCOMPER	Computers & Peripherals	13	4112
ITCELEC	Electronic Equipment	23	323
ITCIT	IT Services	58	2919
ITCNET	Internet Software & Servi	23	6797
ITCSEMIEQP	Semiconductor Equipment	46	6038
ITCSOFT	Software	96	6321
MATCHEM	Chemicals	34	741
MATCNSTPAK	Construction Materials & Packaging	17	214
MATMETXSTL	Metals & Mining ex Steel	6	121
MATPAPER	Forest & Paper Products	1	7
MATSTEEL	Steel	8	92
TELTELECOM	Telecommunication Services	12	809
UTLUTIL	Utilities	55	1383

Table 18: CHNA industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDIAUTOCMP	Automobiles & Components	132	481
CDICONSVCS	Consumer Services	26	47
CDIHHDUR	Household Durables	57	248
CDILEISEQ	Leisure Equipment & Products	8	11
CDIMEDIA	Media	62	117
CDIRETAIL	Retailing	27	67
CDITXTLUX	Textiles Apparel Luxury Goods	46	92
CSTALCTOB	Alcohol & Tobacco	31	493
CSTXALCTOB	Cons Staples x Alcohol & Tob	148	393
EGYENERGY	Energy	58	622
FINBANK	Banks	40	1715
FINDIV	Diversified Financials	56	458
FININS	Insurance	7	371
FINMULTSEC	Multi-Sector Holdings	38	6
FINREALMGT	Real Estate Management & Dev	74	153
HLCHLC	Health Care	310	722
INDBLDPRD	Building Products	23	40
INDCOMPRO	Commercial & Prof Services	54	78
INDCONMACH	Construction Farm Mach Trucks	47	192
INDDEFENG	Aerospace & Def Const & Eng	124	365
INDELECEQ	Electrical Equipment	178	538
INDINDMACH	Industrial Machinery	180	293
INDTRAN	Transportation	85	397
INDTRDDIST	Trading Companies & Distrib	31	62
ITCCOMMEQ	Communications Equipment	55	257
ITCCOMP	Computers & Peripherals	49	144
ITCELECEQO	Elec Eq Instr Comp & Off Elec	201	495
ITCITSERV	IT Services	94	139
ITCSEMICON	Semiconductors	208	727
ITCSOFTNET	Software & Internet Services	113	382
MATAGR	Fertilizers & Agriculture Chem	32	60
MATALU	Aluminum	26	71
MATCHEMIND	Chemicals & Industrial Gases	175	389
MATCONSTM	Construction Materials	26	53
MATDIVPREC	Div & Prec Metals & Mining	81	276
MATPAKPAP	Containers Packaging & Paper	36	45
MATSPECHEM	Specialty Chemicals	56	81
MATSTEEL	Steel	51	122
TELTELECOM	Telecommunication Services	16	154
UTLUTIL	Utilities	97	480

Table 19: INDI industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDIAUTOCMP	Automobiles & Components	56	310
CDIMEDIA	Media	15	15
CDIRETAILSRVC	Consumer Services & Retailing	45	107
CDITXTLEI	Consumer Apparel	40	89
CSTALL	Consumer Staples	62	364
EGYENERGY	Energy	20	327
FINBANK	Banks	43	589
FINDIV	Diversified Financials	81	483
FINMULTSEC	Multi-Sector Holdings	63	41
FINREALMGT	Real Estate Management & Dev	34	102
HLCHLC	Health Care	85	335
INDBLDPRDSRVC	Building Prod & Ind Mach	82	128
INDCONDEF	Aero Def & Ind Cong & Elec Eq	118	424
INDTRANDIST	Transportation & Distrib	28	132
ITCCOMSEMI	Tech Hardware & EQ & Semicon	20	42
ITCITSERV	IT Services & Telecom	53	748
ITCSOFTNET	Software & Internet Services	22	96
MATAGR	Fertilizers & Agriculture Chem	21	44
MATCHEMPAK	Chemicals & Packaging	87	165
MATCONSTM	Construction Materials	25	130
MATDIVMET	Metals & Mining	55	196
UTLUTIL	Utilities	34	264

Table 20: NAHC industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
HLCBIOTECH	Biotechnology	115	859
HLCEQSUPLS	HC Eq Sup Life Sci Tools Svcs	74	1809
HLCMGDHLC	Managed Health Care	9	711
HLCPHARMA	Pharmaceuticals	40	2309
HLCPROVSVC	Health Care Providers & Svcs	54	365

Table 21: SAFR industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDIALL	Retailing	27	56
CONSUALL	Consumer Services	22	72
EGYMAT	Energy and Materials	16	19
FINALL	Financials	20	118
FINREAL	Real Estate	16	11
INDUTL	Industrials and Utilities	17	16
MATGOLDPREC	Mining	9	46

Table 22: BRAZ industry schema

BFRE Industry Code	BFRE Industry Name	# Assets	Mkt Capt ($bn)
CDIALL	Consumer Discretionary	48	24
CSTALL	Consumer Staples	22	73
EGYMAT	Energy and Materials	21	207
FINALL	Financials	50	311
HLCHLC	Health Care	13	18
INDINDUST	Industrials	35	77
MATMIN	Metals and Mining	15	67
TELUTL	Telecom and Utilities	47	149

8.6 BFRE 2.0 Gamma Adjustment Methodology

This section describes the methodology for calculating factor risk gamma scaling and specific risk gamma scaling in BFRE 2.0 models. The gamma γ scaling is designed to enhance the responsiveness of 1.0 EWMA forecasts to the most recently observed volatility conditions. The adjustment parameter is computed with reference to the cross-section of realised volatilities.

Factor risk gamma adjustment

The BFRE 1.0 time series based EWMA factor covariance matrix at time t is scaled by a scaling γ_t, chosen to minimise the mean-squared-error (MSQE) of the EWMA forecast σ_{it}^{EWMA}, where i ranges from 1 to K for K factors in the matrix, with respect to the most recent[30] realised factor volatilities $\sigma_{i,t}^{realised}$:

$$\min_{\gamma_t} \frac{1}{K} \sum_{i=1}^{K} [\sigma_{i,t}^{realised} - \gamma_t \sigma_{i,t}^{EWMA}]^2 \tag{58}$$

where:

- $\sigma_{i,t}^{EWMA}$ is the time series based EWMA forecast for each factor $i = 1, ..., K$ at time t
- The summation includes the Market factor, style factors, core country factors and core industry factors
- For Short Horizon models with 1-month forecast horizon, monthly realised volatility is calculated from daily Factor Returns $f_{i,j}$, for each factor $i = 1, ..., K$:

$$\sigma_{i,t}^{realised} = \sqrt{\sum_{j=t}^{t-20} (f_{i,j})^2} \tag{59}$$

- For Long Horizon models with 6-month forecast horizon, quarterly realised volatility is calculated from daily Factor Returns $f_{i,j}$, for each factor $i = 1, ..., K$:

$$\sigma_{i,t}^{realised} = \sqrt{\sum_{j=t}^{t-63} (f_{i,j})^2} \tag{60}$$

Solving for γ_t gives:

$$\gamma_t = \frac{\sum_{i=1}^{K} \sigma_{i,t}^{realised} \sigma_{i,t}^{EWMA}}{\sum_{i=1}^{K} (\sigma_{i,t}^{EWMA})^2} \tag{61}$$

While γ_t has a time series average of 1, it scales the EWMA estimates up(down) when recent observed realised volatilities are high(low) relative to the EWMA estimates.

Specific risk gamma adjustment

The methodology for calculating specific risk gamma scaling is similar to factor risk gamma and makes use of the asset-specific realised volatilities instead of factor realised volatilities:

$$\gamma_t = \frac{\sum_i \sigma_{i,t}^{realised} \sigma_{i,t}^{EWMA}}{\sum_i (\sigma_{i,t}^{EWMA})^2} \tag{62}$$

where:

[30] most recent monthly realised volatilities for Short Horizon models and quarterly for Long Horizon models

- $\sigma_{i,t}^{EWMA}$ is the time series based EWMA specific risk forecast for each asset i at time t
- For Short Horizon models with 1-month forecast horizon, monthly realised volatility is calculated from daily specific returns $u_{i,j}$, for each asset i in the model estimation universe:

$$\sigma_{i,t}^{realised} = \sqrt{\sum_{j=t}^{t-20} (u_{i,j})^2} \tag{63}$$

- For Long Horizon models with 6-month forecast horizon, quarterly realised volatility is calculated from daily specific returns $u_{i,j}$, for each asset i in the model estimation universe:

$$\sigma_{i,t}^{realised} = \sqrt{\sum_{j=t}^{t-63} (u_{i,j})^2} \tag{64}$$

8.7 BFRE 2.0 Structural Forecast Methodology

This section describes the methodology for the additional adjustment to specific risk gamma scaling to give structural specific risk forecast. The specific risk gamma scaling in the first step is common to all assets while the second-step adjustment described in this section exploits asset-specific characteristics to add additional variations to the common scaling. The structural specific risk forecast is based on a panel regression using realised volatilities, factor exposures and gamma-adjusted specific risk forecast from the previous step.

At each month-end date t, fit realised volatilities using factor exposures, gamma-adjusted specific risk forecast and the interaction between factor exposures and gamma-adjusted specific risk forecast. The panel for the regression is defined over a two-dimensional grid of assets i and dates s:

$$\sigma_{i,s}^{realised} = a + \sum_k b_k X_{i,\tilde{s},k} + c\sigma_{i,\tilde{s}}^{\gamma-adj} + \sum_k d_k X_{i,\tilde{s},k}\sigma_{i,\tilde{s}}^{\gamma-adj} \tag{65}$$

where:

- $\sigma_{i,s}^{realised}$ is the realised volatility of asset i at each month-end dates, $s <= t$
- $\tilde{s}$ is the date given by s minus the lookback period in Table 23
- $X_{i,\tilde{s},k}$ is the factor exposure of asset i to factor k at the start of the months
- $\sigma_{i,\tilde{s}}^{\gamma-adj}$ is the gamma-adjusted specific risk forecast from the previous step
- a, b_k, c, d_k are coefficients estimated from the regression
- The panel regression only includes assets in the core estimation universe at time $\tilde{s}$ for each panel date s
- Short Horizon models include Size, Volatility, Momentum and Liquidity while Long Horizon models include Size and Volatility
- The panel regression uses a rolling window with length given in Table 23

Table 23: BFRE structural specific risk parameters

Model frequency	Lookback period	Rolling window length
Daily, 2.0 Short Horizon	21 days	24 months
Daily, 2.0 Long Horizon	125 days	48 months
Weekly (WRLD and EMKT), 2.0 Short Horizon	4 weeks	24 months
Weekly (WRLD and EMKT), 2.0 Long Horizon	26 weeks	48 months

The structural specific risk forecast at time t (which is not necessarily a month-end) is calculated using the most up-to-date coefficients estimated from the panel regression (from previous month-end), most up-to-date factor exposures and gamma-adjusted specific risk forecast:

$$\sigma_{i,t}^{STR} = \hat{a} + \sum_k \hat{b_k} X_{i,t,k} + \hat{c}\sigma_{i,t}^{\gamma-adj} + \sum_k \hat{d_k} X_{i,t,k}\sigma_{i,t}^{\gamma-adj} \tag{66}$$

where:

- $\sigma_{i,t}^{STR}$ is the structural specific risk forecast at time t

Bibliography

[1] B. Rosenberg, "Extra-Market Components of Covariance in Security Returns," *Journal of Financial and Quantitative Analysis*, vol. 9, no. 2, 1974.

[2] S. Heston and K. Rouwenhorst, "Industry and Country Effects in International Stock Returns," *Journal of Portfolio Management*, vol. 21, no. 3, 1995.

[3] A. J. Miller, *Subset Selection in Regression (Second Edition).* London: Chapman and Hall, 2002.

[4] R. R. Hocking, "Developments in linear regression methodology: 1959-1982," *Technometrics*, vol. 25, pp. 219–230, 1983.

[5] M. L. Thompson, "Selection of variables in multiple regression: Part 1. a review and evaluation. part 2. chosen procedures, computations and examples.," *Int. Statist. Rev.*, vol. 46, pp. 1–19 and 129–146, 1978.

[6] R. Banz, "The Relationship between Return and Market Value of Common Stocks," *Journal of Financial Economics*, vol. 9, no. 1, 1981.

[7] N. Jegadeesh, "Evidence of Predictable Behavior of Security Returns," *Journal of Finance*, vol. 65, no. 3, 1990.

[8] B. Jacobs and K. Levy, "Disentangling Equity Return Regularities: New Insights and Investment Opportunities," *Financial Analysts Journal*, vol. 44, no. 3, 1988.

[9] A. Subrahmanyam, "Distinguishing Between Rationales for Short-Horizon Predictability of Stock Returns," *The Financial Review*, vol. 40, no. 1, 2005.

[10] N. Jegadeesh and S. Titman, "Returns to Buying Winners and Selling Losers: Implications for Stock Market Efficiency," *Journal of Finance*, vol. 48, no. 1, 1993.

[11] J. Lee and D. Stefek, "Do Risk Factors Eat Alphas?," *Journal of Portfolio Management*, vol. 34, no. 4, 2008.

[12] A. Sarr and T. Lybek, "Measuring Liquidity in Financial Markets," *IMF Working Paper*, vol. WP/02/232, 2002.

[13] G. Connor, L. Goldberg, and R. Korajczyk, "Portfolio Risk Analysis," *Princeton University Press*, 2010.

[14] G. Bekaert, C. Harvey, and C. Lundblad, "Liquidity and Expected Returns: Lessons from Emerging Markets," *The Review of Financial Studies*, vol. 20, no. 5, 2007.

[15] Y. Amihud, "Illiquidity and stock returns: cross-section and time-series effects," *Journal of Financial Markets*, vol. 5, no. 1, 2002.

[16] C. Florackis, A. Gregoriou, and A. Kostakis, "Trading Frequency and Asset Pricing on the London Stock Exchange: Evidence from a New Price Impact Ratio," *Journal of Banking and Finance*, vol. 35, no. 21, 2011.

[17] R. Goyenko, C. Holden, and C. Trzcinka, "Do Liquidity Measures Measure Liquidity?," *Journal of Financial Economics*, vol. 92, no. 2, 2009.

[18] K. Daniel and S. Titman, "Market reaction to tangible and intangible information," *Journal of Finance*, vol. 61, 2006.

[19] R. McLean, J. Pontiff, and A. Watanabe, "Share issuance and cross-sectional returns: International evidence," *Journal of Financial Economics*, vol. 94, 2009.

[20] A. Rudd and H. Clasing, *Modern Portfolio Theory: The Principles of Investment Management (Revised second edition)*. Probus Publishing Co., 1988.

[21] R. Grinold and R. Kahn, *Active portfolio management: a quantitative approach for providing superior returns and controlling risk (Second Edition)*. McGraw-Hill, 1999.

[22] A. Timmermann, "Chapter 4 forecast combinations," *Handbook of Economic Forecasting*, vol. 1, pp. 135–196, 2006.

[23] W. Newey and K. West, "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix," *Econometrica*, vol. 55, no. 3, 1987.

[24] A. J. Patton, "Volatility forecast comparison using imperfect volatility proxies," *Journal of Econometrics*, vol. 160, no. 1, pp. 246–256, 2011.

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